Honours Thesis - Simulation of Squeezed Light Generation in an Optical Resonator

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CONTENTS

I. TODO

- Check if Tong notes are a good reference with MacRae
- Derive three-level semi-classical Hamiltonian
- Write section on interferometric cavities (Fabry-Perot, spherical, etc.)
- Implement and write up Jaynes-Cummings Model

II. PROJECT MOTIVATION

MacRae's Outline:
Quantum fluctuations of the electromagnetic field impose a fundamental limit on measurement sensitivity, known as the standard quantum limit (SQL). Exotic quantum optical states known as "squeezed light" can surpass the SQL. Squeezed states are a hot topic in research, particularly in quantum communication, metrology, and optical quantum computing. This project will explore theoretical models predicting strong squeezing when atoms are placed inside an optical cavity. The student will develop and run numerical simulations to investigate the conditions for optimal squeezing, using tools from quantum optics and cavity QED theory. Through this work, the student will gain experience in programming, numerical methods, and the theoretical framework of quantum light–matter interactions.

The past century since the formulation of quantum mechanics has led to unrivaled technological progress, especially in recent years. Methods in material sciences, such as density functional theory for computational modelling of solid state quantum mechanical systems [**?**], have brought commercial transistors down to the scale of nanometers. With greater understanding of and ability to control quantum phenomena, further possibilities become available. Modern avenues for advancing quantum technologies include quantum key distribution, quantum sensing, and quantum computation, to name a few.

Any emerging technology tends to follow the rule ascribed to increasing system complexity sometimes stated as 'more is different' with reference to the 1972 paper by Physicist Philip W. Anderson [**?**]. This especially applies to quantum networks. As greater quantum control is achieved, greater quantum networking capabilities will be required to reap maximal benefits, thereby necessitating the development of improved methods for transmitting quantum information.

One near-term threat which looms over modern encryption methods is the threat of the 'harvest now, decrypt later' information gathering strategy by bad actors [**?**]. Since quantum algorithms like Shor's algorithm [**?**] (or even brute force GPU techniques at massive scale) have the potential to compromise 'securely' transmitted information, finding alternative quantum-secure methods is considered an issue of national defence. Quantum key distribution in which photonic quantum bits encode information has been proposed as one such method [**?**] and could be applied to critical information channels in the near-term given a scalable method for building quantum networks.

Currently, there are two main hurdles in developing large scale quantum networks: creating a continent-spanning network compatible with current methods of producing quantum information; and transmitting quantum information with sufficiently low loss/noise such that the information sent may actually be used. There is potential that the first problem has already been solved, however. Fibre optic networks already span the world, but have a limited transparency window to which information can be transmitted with minimal losses, and current quantum techniques do not operate produce photonic quantum information in this range. So if a method of efficiently producing quantum information with minimal losses at fibre optic wavelengths is found, then progress towards both problems would be obtained.

After this, the problem becomes advancing techniques for minimizing losses in quantum coherence during fibre optic transmission [**?**] and all other forms of efficiency such that amount of information transmitted through the channel may be maximized [**?**].

Progress has been made on forming quantum-optical networks, such as in America where an advanced quantum network architecture has been developed on Long Island [**?**], or in China where a 4600 kilometer land-based quantum network [**?**].

This thesis covers a theoretically viable method for frequency conversion of quantum information encoded in an electromagnetic field (continuous variable quantum information). The technique involves interfering three electromagnetic fields within a sample of Rubidium-87 gas to induce transitions such that one additional field is produced. Rubidium-87 is a widely studied candidate for applications in short-timescale quantum memory [?][?], quantum repeaters [?], and quantum computation [?]. We will focus on a 4-level transition substructure with one transition frequency compatible with modern quantum methods, one transition frequency compatible with telecom frequencies in the conventional band (C-band), and two frequencies which will be leveraged for interference effects in a process known as four wave mixing (FWM) [?].

TODO - Write what the outline of the thesis will be here!!!

III. BACKGROUND INFORMATION

This project involves understanding how hot atomic vapours interact with external electromagnetic fields to induce desirable physical phenomena. To convert frequencies, FWM can be applied with the added bonus that it can generate squeezed light [?], which is useful in continuous variable quantum information applications since optical squeezing causes uncertainty in one quadrature (amplitude or phase) to be reduced while being increased in the other. FWM may be enhanced by electromagnetically induced transparency (EIT) [?], a process in which an atomic medium becomes effectively transparent to a frequency of light near-resonance with one of its transition frequencies under certain conditions. We will tackle these topics in reverse order to build up to the idea of optical squeezing.

In order to solve for signal propagation, we must find the steady state system dynamics via the Lindblad master equation, then use the result to solve for the field propagation through the medium step by step using the Maxwell-Bloch equations. The master equation is a function of field strength, so we can produce a combined solution algorithm to obtain the field strength of the converted frequency field at the end of the sample. This result tells us how much of the signal has been converted via FWM.

A. Electromagnetically Induced Transparency

In order to gain an intuition on EIT, we will start with a classical toy model of the atom. As is a common theme in physics, a hydrogenic atom in a time dependent electromagnetic field can be approximately modelled as a driven harmonic oscillator. So rather than dealing with quantized photons, we will consider a classical oscillating electromagnetic fields with frequencies nearly resonant with those of the atomic transitions.

To do this, a few assumptions will be made [?]: first, the wavelengths of the light sources interacting with the atom have wavelengths much greater than the Bohr radius of the atom ($\lambda \gg a_0$); second, the detunings of incoming beams are small in proportion to the transition frequencies considered. These two assumption happen to be consistent with one another.

The first assumption allows for us to approximate the electromagnetic field experienced by the atom as constant in space but varying in time since the change in electromagnetic field strength between the two ends the atom in the transverse wave direction is relatively small.

There exists the relation between the electric and magnetic field strengths of light which arises from Maxwell's equations

$$B = \frac{E}{c}. \tag{1}$$

The Lorentz force is

$$\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right) \tag{2}$$

where q is charge, and $\vec{v}$ is particle velocity. Using the Bohr model for approximation, we have that quantized electron velocities for a hydrogenic atom have magnitude

$$v_n = \alpha \frac{Z}{n} c \tag{3}$$

where α is the fine structure constant, Z is the atomic number, and n is the principle quantum number [?]. We can therefore see by rough approximation that we have

$$|\vec{v} \times \vec{B}| \lessapprox \alpha Z c \frac{E}{c} = \alpha Z E. \tag{4}$$

We are considering Rubidium with atomic number $Z = 37$, but this neglects the effects of screening caused by the electrons in inner shells of the atom. Electron position is distributed in a cloud about the atom for any orbital, though we can expect the 36 electrons of inner shells between the valence electron (singular in its shell) and the nuclear charge of $37e$ to roughly cancel $36e$ of charge such that the valence electron experiences about $1e$. By this line of reasoning we have

$$|\vec{v} \times \vec{B}| \lessapprox \alpha E. \tag{5}$$

Thus, the force due to the external magnetic field is about 1/137 that of the external electric field. So we may neglect the external magnetic field for simplification.

$$\vec{F} \approx q\vec{E} \tag{6}$$

All of these approximations can be justified from the perspective that this model is a first estimate for system behaviour which will be refined and iterated upon.

This brings us to a physicist's favorite abstraction: the harmonic oscillator. The potential of which the valence electron experiences can be expanded in a Taylor series and approximated as quadratic. The mass of the nucleus is much greater than the electron mass, so we may approximate the nucleus to be fixed. So we have a mass with some equilibrium distance from a fixed point under a quadratic potential, i.e. a simple harmonic oscillator. We may then add in our external time-dependent driving force to obtain a driven harmonic oscillator.

When a charge accelerates, it emits photons as its movement produces a changing electric and magnetic field. So when our system is driven on resonance, then an electromagnetic field of that same frequency will be produced. Here lies the idea behind electromagnetically induced transparency: by driving a cloud of these hydrogenic atoms on resonance, the field essentially 'passes through' the medium as the dipole response of the medium matches the driving frequency. This abstraction is known as the Lorentz oscillator [**?**].

It is helpful to imagine the dipole response of the medium as a vector field coupled to the external driving field of frequency ω. When we quantize the system, this is encapsulated by the expectation value of the dipole moment of the atomic medium $\langle\hat{\vec{d}}\rangle = -e\langle\hat{\vec{r}}\rangle$ since the polarization is given by $\vec{P} = N\langle\hat{\vec{d}}\rangle$, where $\vec{r}$ is the relative position vector and N is the number of atoms in the medium [**?**].

So far we have considered the atomic system to behave with a linear response to the external field, however, this is not the case. The true polarization of the system actually behaves in a nonlinear fashion with respect to the driving field. This response is medium dependent and encapsulates all the interactions of the system, so it can be quite complicated. For this reason, it is often abstractly written in the expanded polynomial form [**?**]

$$\vec{P} = \epsilon_0 \left(\chi^{(1)}\vec{E} + \chi^{(2)}\vec{E}\vec{E} + \chi^{(3)}\vec{E}\vec{E}\vec{E} + \dots \right) \tag{7}$$

where $\chi^{(i)}$ is the i^{th} order electric susceptibility rank $i+1$ tensor. For weak electric fields, the linear approximation may be made to give

$$\vec{P} \approx \epsilon_0 \chi^{(1)}\vec{E} \tag{8}$$

then by $N\langle\hat{\vec{d}}\rangle = \epsilon_0\chi^{(1)}\vec{E}$ we have

$$\chi^{(1)} = \frac{N|\langle\hat{\vec{d}}\rangle|}{\epsilon_0|\vec{E}|}. \tag{9}$$

What frequencies are allowed to pass through the atomic medium is decided by the linear electric susceptibility of the material $\chi^{(1)}$. When light travels through some material with complex electric susceptibility (i.e. complex index of refraction), then the light will receive some phase shift due to the real-valued component of susceptibility $\chi_R^{(1)}$ and some decay due to the imaginary-valued component $\chi_I^{(1)}$, such that the oscillating electric field will propagate as

$$E = E_0 e^{i(kz-\omega t)} e^{i\chi_R^{(1)}kz/2} e^{-\chi_I^{(1)}kz/2} \tag{10}$$

where the rate of decay then grows exponentially with distance traveled z through the medium. The decay term $e^{-\chi_I^{(1)}kz/2}$ is known as Beer-Lambert law [**?**].

So, we can say that when the imaginary component of linear susceptibility goes to zero, the medium becomes 'transparent' to the driving field. This gives us a mathematical way to determine if EIT is present in a system.

One additional phenomenon which happens in tandem with EIT is slowed light. The index of refraction of a material is

$$n = \sqrt{1 + \chi^{(1)}}, \tag{11}$$

and $\chi^{(1)}$ for the Lorentz oscillator may be expressed as [?]

$$\chi^{(1)}(\omega) = \frac{Ne^2}{\epsilon_0 m_e} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \tag{12}$$

where ω_0 is the resonant frequency of the system, ω is the driving frequency, and γ is the decay rate (takes on the role of drag in the oscillator approximation). The group velocity of a wave can be written [?]

$$v_g = \frac{\partial \omega}{\partial k} = \frac{c}{n + \omega \frac{\partial n}{\partial \omega}}. \tag{13}$$

Combining the information from these 3 equations tells us that $\chi^{(1)}$ tends to be very sensitive with respect to the driving frequency near $\omega \approx \omega_0$, so if the driving laser is tuned near resonance such that EIT occurs, then the derivative $\partial_\omega n$ may become sufficiently large such that the group velocity of the propagating electromagnetic field is significantly reduced. Thereby, the dipole response of the medium may result in the propagation of slowed light when EIT is present.

Unfortunately, an atom is not a mass on a spring, but this model gives us a good idea of how EIT works as a first step. In reality, we must consider the structure of atomic states and the interference of transition pathways.

1. *Structure of atomic spectra*

Atomic states for the electron of a hydrogenic atom are enumerated by the quantum numbers n (principle quantum number), l (angular momentum quantum number), m_l (magnetic quantum number), and m_s (spin quantum number) [?].

$$n \in \mathbb{N} \tag{14}$$
$$l \in \{0, 1, \ldots, n-1\} \tag{15}$$
$$m \in \{-l, \ldots, l\} \tag{16}$$

Notably, there is also the electron spin s quantum number which may take on values $s = \pm 1/2$ for hydrogenic atoms.

The energy levels for an unperturbed hydrogenic atom modelled with just the coulomb potential are quite simple as they just depend on the principal quantum number.

$$E_n = -\frac{(Z\alpha)^2 mc^2}{2n^2} = -\frac{Z^2 Ry}{n^2} \tag{17}$$

where Z is the atomic number, α is the fine structure constant, m is the electron mass, and c is the speed of light [?]. These unperturbed energy levels each have degeneracy of n^2. The real picture, is more complicated, however. There are fine structure and hyperfine structure perturbations on these energy levels, splitting energy levels and decreasing their degeneracies.

Fine structure corrections arise due to relativistic corrections to the non-relativistic Schrödinger's equation. In particular, we have a pure relativistic correction from considering electron momentum (which is a non-negligible fraction of the speed of light), a spin-orbit coupling correction, and the Darwin correction [?]. To first order perturbation theory, the combined energy shift of these contributions is

$$\Delta E_{fs} = \frac{(Z\alpha)^4}{2} \frac{mc^2}{n^3} \left(\frac{3}{4n} - \frac{2}{2j+1} \right) \tag{18}$$

where $j = |l \pm s| = |l \pm 1/2|$ since we have $s = 1/2$ for hydrogenic atoms [?]. Notice that these perturbations are scaled by a factor of $(Z\alpha)^2$ in comparison to Eq. (??) such that for atoms with a sufficiently large atomic number, these approximate energy shifts arrived at by perturbation theory break down.

There exist yet smaller energy shifts due to the interaction between the internal magnetic fields of the nucleus and the orbital motion of the electron along with its spin [?]. The hyperfine energy splitting term is given as

$$\Delta E_{hfs} = \frac{A}{2}\left[F(F+1) - I(I+1) - j(j+1)\right] \tag{19}$$

where A is known as the hyperfine constant, F is the total angular momentum quantum number with operator $\hat{F} = \hat{I} + \hat{S}$, I is the nuclear spin quantum number, and j is defined as before [?][?].

Spectroscopic notation is used to describe atomic levels. It is expressed

$$n^{2s+1}L_j \tag{20}$$

where n, s, and j are atomic numbers as we have defined, $2S+1$ is the multiplicity of states, L is the atomic term symbol $(S, P, D, F, \ldots)$ [?]. Notice that for a hydrogenic atom we will always have multiplicity of $2s+1 = 2$ which correspond to the 2 hyperfine states $F = 1$ and $F = 2$

2. *Atomic transitions*

In a hydrogenic atom, the transitions between states are mediated by excitations in the electromagnetic field (i.e. photons). It is crucial in atomic physics to understand the probabilities of these atomic state transitions.

The interaction between an atom and the electromagnetic field is not a linear interaction, creating difficulty in finding an exact solution. To obtain approximate behaviour, first order perturbation theory can be applied to dominating effects of this interaction [?]. From this, one obtains the transition dipole matrix with elements

$$\vec{\mu}_{ji} = \langle j|\hat{\vec{d}}|i\rangle = e\langle j|\hat{\vec{r}}|i\rangle \tag{21}$$

where $\hat{\vec{d}} = e\hat{\vec{r}}$ is the dipole transition operator and $\hat{\vec{r}}$ is the position operator [?].

If $\vec{\mu}_{ji} = 0$, then the transition between states $|i\rangle$ and $|j\rangle$ is a 'forbidden' dipole transition, otherwise the transition is an allowed dipole transition. This criterion enforces the dipole selection rules that [?]

$$\Delta l = \pm 1 \tag{22}$$

$$\Delta m_l = 0, \pm 1 \tag{23}$$

$$\Delta F = 0, \pm 1 \tag{24}$$

$$F = 0 \leftrightarrow F = 0 \text{ is forbidden.} \tag{25}$$

It is important to note that these transition rules are not always perfectly enforced. Since perturbation theory was applied to obtain them, there exist transitions which disobey these rules, however, such transitions occur with significantly decreased probability.

3. *Three-level atoms*

In order to meaningfully utilize the power of quantum information via quantum states, it is beneficial to work with simplified, yet somewhat abstract, systems. It would be ideal if we could realize any imaginary quantum systems that exactly match our mathematical descriptions. Reality is not so convenient, however. We must work within the constraints enforced by the available quantum systems in terms of allowed states and transitions between those states.

With enough ingenuity, we can use existing systems to approximate useful mathematical abstractions. Enter, the three-level (hydrogenic) atom. All hydrogenic atoms, in theory, have infinitely many states up to ionization ($n \in \mathbb{N}$), and transition probabilities between many of these states can be non-zero. Though by finding a subset of states with desired characteristics and leveraging transition rules where any undesired state transitions are suppressed, a physical atom can be used to approximate a model system. This becomes especially useful in the limit of a large number of atoms where, statistically, the desired transitions are the most probable, and any undesired transitions are vastly outnumbered, such that the system tends towards the desired behaviour in the limit of many particles.

A three-level atom has three distinct states with some set of allowed transitions between those states. There exist Λ-systems, V-systems , and Ξ-systems, each depicted in Fig. **??**. An common example of a Λ-system exists in ^{87}Rb with the collection of hyperfine split states $5^2S_{1/2}$ for $F = 1$ and $F = 2$ with energy differece given according to (**??**)

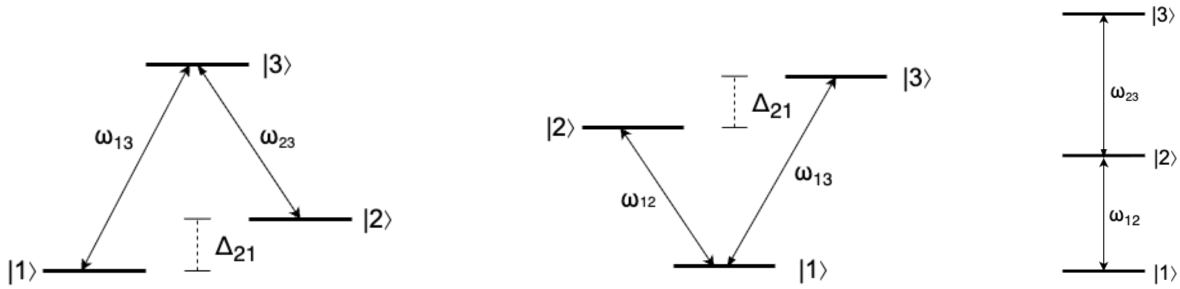

FIG. 1: Three-level atomic configurations: (left) Λ-system, (middle) V-system, and (right) Ξ-system.

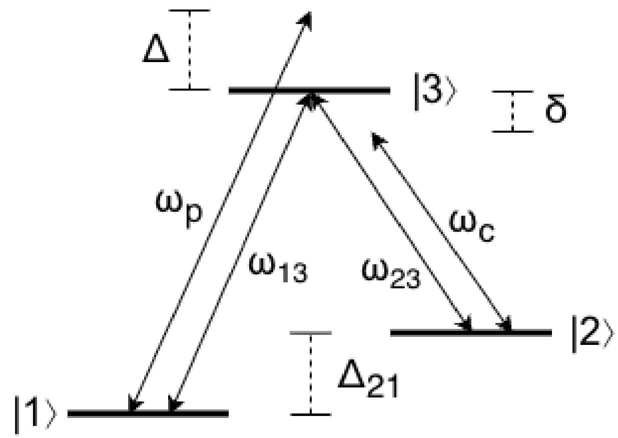

FIG. 2: A thee-level Λ-system with natural transition frequencies ω_{13} and ω_{23} which is driven near-resonance by detuned beams operating at ω_p and ω_c

so that they take on the roles of states $|1\rangle$ and $|2\rangle$, respectively, and state $5^2P_{1/2}$ for $F = 1$ acts as the excited state $|3\rangle$ [**?**].

Some points of note in this system are as follows. State $5^2S_{1/2}$ $F = 1$ is the absolute ground state of the system. The nuclear spin of ^{87}Rb is 3/2 [**?**], so states $|1\rangle$ and $|2\rangle$ differ by the direction of electron spin $s = \pm 1/2$, so the energy difference between these states is the result of hyperfine splitting. State $|3\rangle$ has the same principle quantum number of $n = 5$ as states $|1\rangle$ and $|2\rangle$ but a different angular momentum quantum number, indicating that this state is due to fine structure (with a hyperfine energy correction). Since states $|1\rangle$ and $|2\rangle$ have the same l value, the transition between these levels is forbidden as per eqs. (**??**).

Of course, a transition could place the electron into some state outside this state subset (for example, the $5^2P_{1/2}$ $F = 2$ state), so to promote population only between these states driving lasers tuned near the transition frequencies are shone on the system to induce only the desired transitions. Then in the limit of Avogadro's number of atoms the desired system behaviour becomes statistically enforced.

In this Λ-system example, the transition amplitude between states $|1\rangle$ and $|2\rangle$ is suppressed, so if the system is in state $|2\rangle$, then in order to transition back to state $|1\rangle$ the system will likely need to transition through the excited state $|3\rangle$. This makes $|2\rangle$ a quasi-stable state. If there were a way to suppress all transitions to state $|3\rangle$, then we would obtain a system in which the atom is bound to states $|1\rangle$ and $|2\rangle$ such that any photons which would stimulate a transition to the excited state cannot be absorbed.

There is, in fact, a way to do just this. By driving the system with external electromagnetic fields (lasers) at the transition frequencies, destructive quantum interference between the two fields along with specific initial conditions results in zero probability for the system to be in the excited state [**?**]. This state in which there is no probability of occupying $|3\rangle$ is then called the 'dark state'. In practice, since perfect precision in real-world systems is difficult, both lasers are detuned from the exact transition frequency. The probe laser stimulates transition 13 $\omega_p = \omega_{13} + \Delta$, $\Delta \ll \omega_{13}$, and the control laser stimulates transition 23 $\omega_c = \omega_{23} + \delta$, $\delta \ll \omega_{23}$ (see Fig. **??**).

The main idea is that the system then becomes effectively 'transparent' to the probe and control laser frequencies. This is how EIT can be produced in practice.

EIT can also be achieved in a Ξ-system of rubidium between the $|1\rangle = 5^2S_{1/2}$, $|2\rangle = 5^2P_{3/2}$, and $|3\rangle = 5^2D_{5/2}$

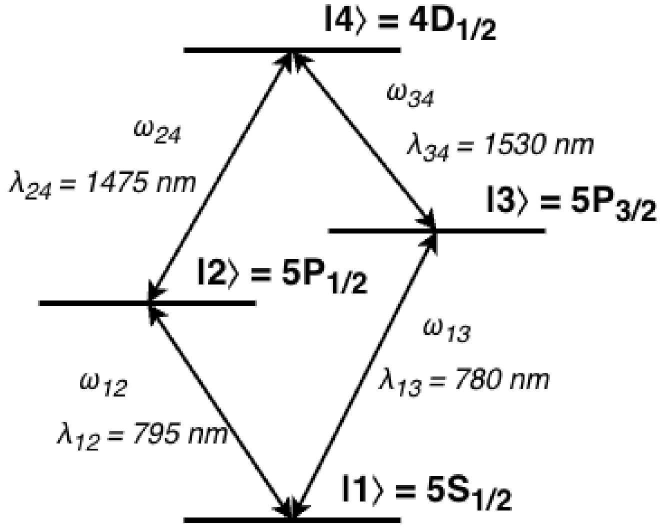

FIG. 3: A diamond configuration atomic system with non-degenerate energy levels $|1\rangle, |2\rangle, |3\rangle, |4\rangle$. The system has allowed transitions $|1\rangle \leftrightarrow |2\rangle$, $|1\rangle \leftrightarrow |3\rangle$, $|2\rangle \leftrightarrow |4\rangle$, and $|3\rangle \leftrightarrow |4\rangle$ with respective energies $\hbar\omega_{12}$, $\hbar\omega_{13}$, $\hbar\omega_{24}$, and $\hbar\omega_{34}$ and corresponding wavelengths λ_{12}, λ_{13}, λ_{24}, and λ_{34}.

states by resonantly driving the $|1\rangle \leftrightarrow |2\rangle$ and $|2\rangle \leftrightarrow |3\rangle$ transitions such that transition pathways interfere to induce transparency [**?**]. This is sometimes called 'cascade' EIT.

EIT in V-systems is less widely applied and studied in comparison to Λ and cascade EIT since there is no stable dark state in V-systems [**?**]. Since two excited states may decay to ground, decoherence effects decrease the efficiency of producing EIT.

4. Diamond configuration four-level atoms

In order to convert electromagnetically encoded quantum information to telecom wavelengths we require a structure with a transition wavelength in the telecom band. Once candidate of note is the diamond transition structure of ^{87}Rb depicted in Fig. **??**. This transition structure can be through of like 2 parallel Ξ-systems which meet at the $|1\rangle$ and $|4\rangle$ states, or as a V-system connected with a Λ-system sharing states $|2\rangle$ and $|3\rangle$.

It has been shown that high efficiency frequency conversion to the original band (O-band) using EIT is possible with an alternative diamond configuration transition structure [**?**]. Chen et al. report that cascade-type and V-type EIT is present in their diamond configuration system, resulting in higher efficiency conversion.

We are instead concerned with conversion to the C-band of range 1530 nm to 1565 nm [**?**]. Henc, we consider the alternative diamond configuration transition structure depicted in Fig. **??**, of which, the $|2\rangle \leftrightarrow |4\rangle$ transition has a wavelength within the C-band. Applying slight detunings to the driving lasers can allow the conversion wavelength to be slightly altered such that minimal losses may be achieved within the C-band window.

5. Electromagnetically induced transparency in hot vapours versus ultracold systems

Consider a system composed of Avogadro's number of three-level or four-level atoms. In the ideal scenario, all atoms would have the exact same transition frequencies such that when the probe and control beams are fired through the cloud. For hot atomic vapours, this is not the case. When the system is in a gaseous excited state: the electromagnetic fields of nearby atoms may interact to induce Stark and Zeeman shifts in their spectra, atoms can collide resulting in state decoherence, and atoms can experience Doppler shifts in the observed frequency of light resulting in a distribution of Doppler shifts throughout the medium in a process known as Doppler broadening. In addition, the laser used will have some bandwidth of emission frequencies. These effects combine to cause spectral broadening such that the transparency window allowed by EIT is suppressed [**?**].

However, it is known that in Λ-systems composed of 'hyperfine-split metastable states' (as described in Section **??**) are unaffected by Doppler broadening if the two applied fields are copropagating [**?**]. This is because for a copropagating pump and control beam (i.e. beams travelling in opposite directions) the Doppler shift observed by any one atom with respect to the pump is equal and opposite to that of the control beam, so they cancel out. This type of system is called Doppler insensitive.

Improved behaviour is achievable for an ultracold atomic system where temperatures are cooled to near-zero-kelvin in order to minimize [**?**]. This is possible with advanced hardware such as a magneto optical trap [**?**], however deploying such technology is currently costly and labor-intensive, so application to large scale networks is not economically viable in the near-term. Therefore, for telecom transmission purposes, it would be favorable to deploy hot atomic frequency converters as opposed to ultracold ones.

If it can be shown that a hot atomic vapour composed of four-level diamond-configuration atoms can be made to behave as Doppler insensitive, then this would be a step towards actualizing a hot C-band frequency converter for electromagnetically encoded quantum information. This is because EIT can be applied to improve FWM efficiency [**?**], and FWM can be used to generate squeezed light [**?**].

B. Four Wave Mixing

FWM is a nonlinear optical process in which three electromagnetic fields of distinct frequencies interact with some medium to produce quantum coherence and interference effects such that an additional distinct field is produced [**?**]. In an atomic medium, FWM is a $\chi^{(3)}$ process [**?**] in which energy conservation and momentum conservation are upheld [**?**].

$$\hbar\omega_a + \hbar\omega_b = \hbar\omega_c + \hbar\omega_d \tag{26}$$

$$\vec{k}_a + \vec{k}_b = \vec{k}_c + \vec{k}_d \tag{27}$$

This allows FWM to be used for optical field frequency conversion. We will be considering FWM in a diamond-configuration atom (see Fig. **??**) where transitions $|1\rangle \leftrightarrow |2\rangle$, $|1\rangle \leftrightarrow |3\rangle$, and $|2\rangle \leftrightarrow |4\rangle$ are driven by near-resonant beams to produce the emission of photons via the $|3\rangle \leftrightarrow |4\rangle$ transition, as previously discussed.

For an atomic vapour, the $\chi^{(2)}$ term is zero valued such that the polarization becomes

$$\vec{P} \approx \epsilon_0 \chi^{(1)} \vec{E} + \epsilon_0 \chi^{(3)} \vec{E}\vec{E}\vec{E} \tag{28}$$

and with three driving fields we have

$$\vec{E} = \vec{E}_a e^{-i\omega_a t} + \vec{E}_b e^{-i\omega_b t} + \vec{E}_c e^{-i\omega_c t} + c.c. \tag{29}$$

such that the nonlinear third order polarization response of the medium becomes

$$\vec{P}^{(3)} = \epsilon_0 \chi^{(3)} \left(\vec{E}_a e^{-i\omega_a t} + \vec{E}_b e^{-i\omega_b t} + \vec{E}_c e^{-i\omega_c t} + c.c. \right)^3 \tag{30}$$

and so it involves the term

$$\epsilon_0 \chi^{(3)} \left(\vec{E}_a \vec{E}_b \vec{E}_c^* e^{-i(\omega_a + \omega_b - \omega_c)t} + c.c. \right) \tag{31}$$

so by the conservation equations Eqs. (**??**) the polarization response of the medium produces a wave of frequency ω_d and wavevector $\vec{k}_d$ [**?**]. Of course, there are other terms involved in the expansion as well. For example, consider term $\chi^{(3)} \vec{E}_a \vec{E}_b \vec{E}_c e^{-i(\omega_a+\omega_b+\omega_c)t}$, which generates a polarization response at frequency $\omega_a + \omega_b + \omega_c$ in a process known as sum-frequency generation, or term $\chi^{(3)} \vec{E}_a \vec{E}_a \vec{E}_a e^{-i3\omega_a t}$ which produces a field at frequency $3\omega_a$ in a process called third-harmonic generation [**?**].

So why is the ω_d field produced so important? What about all the other fields? Well, a photon is emitted when a charge accelerates, so for an electromagnetic field to be produced the valance electron must transition between states. By cleverly selecting the level structure and driving beam frequencies, many undesired transitions pathways can be suppressed and a transition near frequency ω_d can be driven [**?**]. Rogue transitions may produce photons outside the set of frequencies $\{\omega_a, \omega_b, \omega_c, \omega_d\}$, but their number will be negligible in comparison to that of the ω_d field.

For the four-level diamond configuration atom of consideration (see Fig. **??**), driving fields of transition frequency ω_{12}, ω_{13}, and ω_{24} will be applied to induce a near-ω_{34} transition within the telecom C-band frequency range.

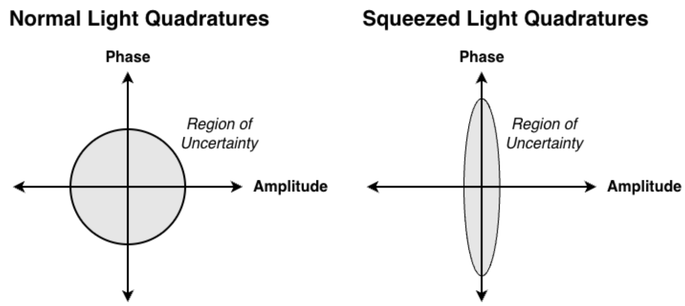

FIG. 4: Cartoon depiction of optical squeezing in the conjugate phase and amplitude quadratures with (left) unmodified light and (right) squeezed light.

1. *Squeezed states*

A significant property of FWM is that the additional field generated can be 'squeezed'. Optical squeezing gives a way to decrease the noise (i.e. loss in quantum information) of the resulting beam, making it a crucial factor to consider. A frequency converter which produces a signal with excessive noise is not useful in quantum information applications. According to Heisenberg's uncertainty principle, the product of the standard deviations of two canonically conjugate observables, $\hat{x}$ and $\hat{p}$ where $[\hat{x}, \hat{p}] = i\hbar$, must satisfy

$$\Delta\hat{x}\Delta\hat{p} \geq \frac{\hbar}{2}. \tag{32}$$

For light of sufficient intensity (i.e. for sufficiently many photons), one can consider its amplitude and phase as canonically conjugate variables [**?**]. For a generically produced beam, one would expect the standard deviations of amplitude and phaseto to be roughly equal. If measurement were to be performed in a single quadrature (amplitude for example), then reducing the uncertainty in that quadrature would result in a more precise overall reading.

When quantum information is stored in the amplitude and phase quadratures of light, it is called continuous variable quantum information [**?**]. Continuous variable measurements are typically performed with homodyne detectors, compare some input beam against a controlled probe beam and takes the difference between their readings [**?**].

This leads to the idea of a squeezed state in which one quadrature has its uncertainty decreased at the expense of the other, such that the uncertainty principle is still observed. It has been shown that optical squeezed states can be produced through a FWM process in optical cavities [**?**][**?**].

The operators for amplitude $\hat{X}$ and phase $\hat{P}$ are given as

$$\hat{X} = \frac{1}{2}\left(\hat{a} + \hat{a}^{\dagger}\right) \tag{33}$$

$$\hat{P} = \frac{1}{i2}\left(\hat{a} - \hat{a}^{\dagger}\right) \tag{34}$$

with associated uncertainties

$$\Delta\hat{X} = \frac{1}{2}\sqrt{1 + 2\delta\left(\hat{a}^{\dagger}\hat{a}\right) + 2Re\left(\delta\left(\hat{a}\hat{a}\right)\right)} \tag{35}$$

$$\Delta\hat{P} = \frac{1}{2}\sqrt{1 + 2\delta\left(\hat{a}^{\dagger}\hat{a}\right) - 2Re\left(\delta\left(\hat{a}\hat{a}\right)\right)} \tag{36}$$

where $\delta\left(\hat{x}\hat{y}\right) = \left\langle\hat{x}\hat{y}\right\rangle - \left\langle\hat{x}\right\rangle\left\langle\hat{y}\right\rangle$ [**?**]. This tells us that the amount of squeezing produced between quadratures is a function of $Re\left(\delta\left(\hat{a}\hat{a}\right)\right) = Re\left(\left\langle\hat{a}\hat{a}\right\rangle - \left\langle\hat{a}\right\rangle\left\langle\hat{a}\right\rangle\right)$.

There are actually two forms of squeezing: single-mode and two-mode squeezing. Single-mode squeezing is typically the result of a phase relationship between emitted photon pairs of the same frequency, while in two-mode squeezing the two photons produces are of different frequencies such that there are correlations between the phase and amplitude of the two beams [**?**]. We are specifically concerned with single-mode squeezed states, and so any mention of squeezing will be in reference to single-mode squeezing.

2. FWM and EIT

FWM and EIT tend to be regarded as complementary phenomena since EIT can enhance FWM [?]. If transparency is induced for the generated transmission field, then it can coherently propagate through the medium with minimal losses. In the presence of both phenomena, the polarization response of the medium would oscillate at the desired fourth frequency to coherently produce a field. In theory, quantum information imposed onto one of the input beams could be coherently absorbed by the medium and re-applied to its response field (this idea is also used in rubidium quantum memories) [?].

One dimension of consideration is as follows. Light is quantized, and any beam containing quantum information is typically weak. Individual photons arrive at the detector in a Poisson process, resulting in shot noise for the detector [?]. The random rate at which photons arrive at the detector causes signal noise. It would be highly desirable for the input beam to follow sub-Poissonian statistics (with the same mean as a Poisson distribution but a lower variance) [?] and for the beam to display photon anti-bunching (arrival events are distributed more evenly as opposed to having intensity spikes) [?].

It has been shown that applying cavity EIT to three-level atoms can produce sub-Poissonian light [?], and that applying FWM to three-level systems can produce photon anti-bunching effects [?]. Combining these techniques theoretically allows for the information of continuous variable beams to be more reliably transmitted to some measurement device, such that readings fall below the shot-noise level.

This makes the combination of FWM and EIT processes in hot atomic vapour composed of four-level diamond configuration atoms an attractive possibility. If achievable, then its implementation could lead to an efficient frequency conversion and transmission method for quantum information.

C. Quantum Methods

1. Master equation

For any quantum system, the total amount of information must be conserved, yet when dealing with atomic systems states can decay and the corresponding photon can be lost to the environment. This type of loss must be accounted for and is not included in typical quantum mechanical descriptions of systems unless the entire environment (i.e. the rest of the universe) is also described. This information would all be contained in the von Neumann density matrix time evolution equation

$$\frac{d\hat{\rho}_T}{dt} = -i\left[\hat{H}_T, \hat{\rho}_T\right] \tag{37}$$

where $\hat{\rho}_T$ is the total system density matrix and $\hat{H}_T$ is the total Hamiltonian [?]. Needless to say, solving this is not an option. By tracing out the environment to isolate the subsystem $\hat{\rho}_S = Tr_E\left[\hat{\rho}_T\right]$, applying the Born approximation where all interactions between the system and environment are assumed to be weak such that they may be traced out for simplification, applying the Markov approximation assuming that the system is memoryless (past states do not affect future states), then finally applying the rotating wave approximation where rapidly evolving components are averaged out and neglected (since time evolution can be factored in terms of complex exponentials $e^{-i\omega t}$) such that the resulting density matrix is strictly non-negative [?][?]. Additional simplification yields the Lindblad master equation as

$$\frac{d\hat{\rho}_S}{dt} = -i\left[\hat{H}_S, \hat{\rho}_S\right] + \sum_{\alpha} \gamma_\alpha \left(\hat{L}_\alpha \hat{\rho}_S \hat{L}_\alpha^\dagger - \frac{1}{2}\left\{\hat{L}_\alpha^\dagger \hat{L}_\alpha, \hat{\rho}_S\right\}\right) \tag{38}$$

where $\hat{L}_\alpha$ is the α^{th} dissipative operator (or jump operator), γ_α is the corresponding decay rate, and $\left\{\hat{a}, \hat{b}\right\} = \hat{a}\hat{b} + \hat{b}\hat{a}$ is the anti-commutator [?].

By solving for $d\hat{\rho}_S/dt = 0$ one may find the steady state density matrix of the system. For large systems this is typically done via computational methods. We are interested in the steady state regime as we will assume that the timescale of laser pulses is much greater than the transient response time of the medium. For short laser pulses, considering transient dynamics is important [?], but as a first step towards frequency conversion of continuous variable quantum information longer pulses should be used. With refinement, shorter pulse durations may be achievable, and then transient dynamics would become relevant, but that is a future engineering challenge.

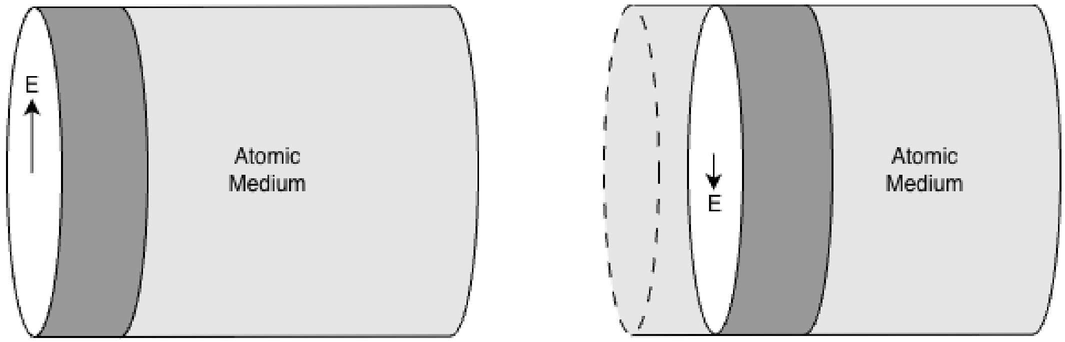

FIG. 5: (Left) the electric field experienced by a point within a region of an atomic medium and (right) a different electric field experienced by a neighboring region.

2. Maxwell-Bloch equations

There is one problem with solving for the steady state density matrix of an atomic sample alone: it neglects propagation effects [**?**]. Even though the entire system may be in a steady state that does not mean that the field strengths are uniform throughout. Any neighboring regions within the sample may experience different fields. Assume that the system can be broken up into cross-sectional regions which all experience approximately the same field strength, which would be the case if the beams were co-propagating or counter-propagating plane waves under the paraxial approximation [**?**]. Then one could consider the field strength of neighboring slices, see Fig. **??**.

The Maxwell-Bloch equations can be derived and applied to account for propagation through the medium. Under the assumptions that the medium is non-magnetic, has no free charges, and no currents due to free charges, we may use Maxwell's equations to obtain the expression

$$\nabla \times \nabla \times \vec{E} + \frac{n^2}{c^2}\frac{\partial^2}{\partial t^2}\vec{E} = -\mu_0 \frac{\partial^2}{\partial t^2}\vec{P} \tag{39}$$

where n is the index of refraction [**?**][**?**]. We may express $\nabla \times \nabla \times \vec{E} = \nabla\left(\nabla \cdot \vec{E}\right) - \nabla^2 \vec{E}$, and apply the slowly varying amplitude approximation such that

$$\vec{E} = \vec{\varepsilon}(z,t) e^{-i(\omega t - kz)} + c.c. \tag{40}$$

where the electromagnetic amplitude envelope $\vec{\varepsilon}(z,t)$ varies in time and space much slower than the carrier wave such that [**?**][**?**]

$$\left|\frac{\partial^2}{\partial t^2}\vec{\varepsilon}(z,t)\right| \ll \left|\omega \frac{\partial}{\partial t}\vec{\varepsilon}(z,t)\right| \tag{41}$$

$$\left|\frac{\partial^2}{\partial z^2}\vec{\varepsilon}(z,t)\right| \ll \left|k \frac{\partial}{\partial z}\vec{\varepsilon}(z,t)\right| \tag{42}$$

$$\tag{43}$$

In this regime, the $\nabla\left(\nabla \cdot \vec{E}\right)$ term then becomes negligibly small such that we may write [**?**]

$$\left(\nabla^2 - \frac{n^2}{c^2}\frac{\partial^2}{\partial t^2}\right)\vec{E} = \mu_0 \frac{\partial^2}{\partial t^2}\vec{P}. \tag{44}$$

Then since we are assuming plane wave propagation in the paraxial approximation we have

$$\left(\frac{\partial^2}{\partial z^2} - \frac{n^2}{c^2}\frac{\partial^2}{\partial t^2}\right)\vec{E} = \mu_0 \frac{\partial^2}{\partial t^2}\vec{P}. \tag{45}$$

By evaluating our derivatives we find that

$$\frac{\partial^2}{\partial t^2}\vec{E} = \left(-2i\omega\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) + \frac{\partial^2}{\partial t^2}\vec{\varepsilon}(z,t) - \omega^2\vec{\varepsilon}(z,t)\right) e^{-i(\omega t - kz)} + c.c. \tag{46}$$

$$\frac{\partial^2}{\partial z^2}\vec{E} = \left(2ik\frac{\partial}{\partial z}\vec{\varepsilon}(z,t) + \frac{\partial^2}{\partial z^2}\vec{\varepsilon}(z,t) - k^2\vec{\varepsilon}(z,t)\right) e^{-i(\omega t - kz)} + c.c. \tag{47}$$

and by the slowly varying amplitude approximation we have that the $\partial_t^2\vec{\varepsilon}(z,t)$ $\partial_z^2\vec{\varepsilon}(z,t)$ terms are negligible. So we have

$$\left(2ik\frac{\partial}{\partial z}\vec{\varepsilon}(z,t) - k^2\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} - \frac{n^2}{c^2}\left(-2i\omega\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) - \omega^2\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} + c.c. = \mu_0\frac{\partial^2}{\partial t^2}\vec{P}$$
$$\left(\frac{2in^2\omega}{c^2}\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) + 2ik\frac{\partial}{\partial z}\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} + \left(\frac{n^2\omega^2}{c^2} - k^2\right)\vec{\varepsilon}(z,t)e^{-i(\omega t-kz)} + c.c. = \mu_0\frac{\partial^2}{\partial t^2}\vec{P}.$$

The phase velocity of light may be expressed as

$$v_p = \frac{\omega}{k} = \frac{c}{n} \tag{48}$$

so we have the relation

$$k = \frac{n\omega}{c} \tag{49}$$
$$\frac{n^2\omega^2}{c^2} - k^2 = 0 \tag{50}$$

thus our expression simplifies to

$$\left(\frac{2ink}{c}\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) + 2ik\frac{\partial}{\partial z}\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} + c.c. = \frac{1}{\epsilon_0 c^2}\frac{\partial^2}{\partial t^2}\vec{P}$$
$$\left(\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) + \frac{c}{n}\frac{\partial}{\partial z}\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} + c.c. = -i\frac{1}{2nk\epsilon_0 c}\frac{\partial^2}{\partial t^2}\vec{P}$$

We can then also apply the slowly varying envelope approximation to $\vec{P}$ such that

$$\vec{P} = \vec{\pi}(z,t)e^{-i(\omega t-kz)} + c.c. \tag{51}$$

$$\frac{\partial^2}{\partial t^2}\vec{P} = \left(-2i\omega\frac{\partial}{\partial t}\vec{\pi}(z,t) + \frac{\partial^2}{\partial t^2}\vec{\pi}(z,t) - \omega^2\vec{\pi}(z,t)\right)e^{-i(\omega t-kz)} + c.c. \tag{52}$$

then we have that

$$\left|\frac{\partial^2}{\partial t^2}\vec{\pi}(z,t)\right| \ll \left|\omega\frac{\partial}{\partial t}\vec{\pi}(z,t)\right| \ll \left|\omega^2\vec{\pi}(z,t)\right| \tag{53}$$

so we may neglet the other terms to obtain

$$\frac{\partial^2}{\partial t^2}\vec{P} \approx -\omega^2\vec{\pi}(z,t)e^{-i(\omega t-kz)} + c.c. \tag{54}$$

Thus, we may write

$$\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) + v_p\frac{\partial}{\partial z}\vec{\varepsilon}(z,t) = i\frac{\omega^2}{2nk\epsilon_0 c}\vec{\pi}(z,t).$$

All vectors will be aligned in an isotropic gaseous media and $k = n\omega/c$, so we can write

$$\frac{\partial}{\partial t}\varepsilon(z,t) + v_p\frac{\partial}{\partial z}\varepsilon(z,t) = i\frac{\omega}{2n^2\epsilon_0}\pi(z,t)$$
$$\frac{1}{v_p}\frac{\partial}{\partial t}\varepsilon(z,t) + \frac{\partial}{\partial z}\varepsilon(z,t) = i\frac{1}{v_p}\frac{\omega}{2n^2\epsilon_0}\pi(z,t)$$
$$\frac{1}{v_p}\frac{\partial}{\partial t}\varepsilon(z,t) + \frac{\partial}{\partial z}\varepsilon(z,t) = i\frac{\omega}{2n\epsilon_0 c}\pi(z,t). \tag{55}$$

This can be simplified even further by moving to a co-moving frame via the transformation $\zeta = z$, $\tau = t - z/v_p$ to obtain [**?**]

$$\frac{\partial}{\partial \zeta}\varepsilon(\zeta,\tau) = i\frac{\omega}{2n\epsilon_0 c}\pi(\zeta,\tau) \tag{56}$$

3. Combined Solution Algorithm

With this, all that remains is finding the dipole response of the system in question. We can use the master equation in tandem with the Maxwell-Bloch equations to solve for the steady state density matrix, then find the expected polarization, then find the electric field strength at the next cross-sectional region. This process can be performed iteratively throughout the medium via computational methods in order to find the desired field strength at the end of the medium. For an idler field, this could then tell us how much field intensity is produced as output via FWM. Due to the use of a co-moving frame, no time evolution is required. This all comes together in Alg. **??**

Algorithm 1 Lindblad-Maxwell-Bloch Algorithm

1: Initialize the electric field strength, frequency, and propagation direction for each field at location $\zeta = 0$ along the atomic medium. Also initialize step size $\Delta\zeta$ and sample length Z (stopping condition).
2: Solve the Lindblad master equation Eq. (**??**) for the steady state density matrix satisfying $d\hat{\rho}/dt = 0$.
3: Use the steady state density matrix $\hat{\rho}$ to find the expected polarization amplitude required in Eq. (**??**) and use it to update the relevant electric field at $\zeta + \Delta\zeta$ as $\varepsilon \to \varepsilon + \partial_\zeta \varepsilon$ via approximation.
4: Update $\zeta \to \zeta + \Delta\zeta$.
5: Repeat steps 2 through 4 until $\zeta \geq Z$.
6: Return ε and $\hat{\rho}$.

IV. THE FIRST STEPS - LEARNING TO USE QUTIP

QuTiP, standing for quantum toolbox in python, is an open source library for simulating quantum systems [**?**]. Our main uses of this library will be for creating operators of the `Qobj` class in the form of finite-dimensional tensors, applying the Lindblad master equation to find the steady state density matrix satisfying $\partial_t \hat{\rho} = 0$ via the `steadystate` function, and taking the expectation value of various operators $\langle \hat{A} \rangle = Tr(\rho \hat{A})$ via the `expect` function.

In order to work up to simulating a four-level atom with multiple Fock bases for photons of distinct frequency modes with dissipative Lindbladian operators, we will begin with a simpler model and reproduce known results, then gradually increase complexity.

A. The Jaynes-Cummings Model

In the Jaynes-Cummings model we consider a two-level atom within a resonant cavity (see Fig. **??**) [**?**]. The field of a single photon within the cavity is quite significant as it will bounce back and forth between cavity walls, building up a non-negligible field. So we must quantize photons in the Fock basis. The cavity is finely tuned to a resonant frequency ω_c such that any other frequency of light destructively interferes with itself. This leads to a simplification for our Fock basis which may be restricted to photons of a single frequency. There can be a detuning between the frequencies, but we will consider the case of perfect resonance here $\omega = \omega_a = \omega_c$.

The states of our system live within the combined Hilbert space $\mathcal{H}_{atom} \otimes \mathcal{H}_{photon}$, and so states are described by $|atom\rangle \otimes |photons\rangle$ where $|atom\rangle$ has two states and $|photon\rangle$ describes all n-photon states. Consider a state containing N photons worth of energy. Then the possible states of the system are $|\downarrow\rangle \otimes |N\rangle$ and $|\uparrow\rangle \otimes |N-1\rangle$.

The Hamiltonian describing this system is given by [**?**] [**?**]

$$\hat{H}_{JC} = \hat{H}_{Atom} + \hat{H}_{Photon} + \hat{H}_{Interaction}. \tag{57}$$

The atomic term is given by

$$\begin{aligned}\hat{H}_{Atom} &= \frac{\hbar\omega}{2}\hat{\sigma}_z \\ &= \frac{\hbar\omega}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.\end{aligned} \tag{58}$$

The photonic term is given by

$$\hat{H}_{Photon} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right).$$

FIG. 6: Depiction of the system for a Jaynes-Cummings model with (left) a resonant cavity containing a single atom in the presence of a driving electromagnetic field resonant with the cavity frequency of ω, with (right) the atomic system depicted as a two-level atom of transition frequency ω_a.

where we may shift by the zero-point energy since it is a constant term to obtain

$$\hat{H}_{Photon} = \hbar\omega\hat{a}^\dagger\hat{a}$$

Then finally, the interaction term is given by

$$\begin{aligned} \hat{H}_{Interaction} &= \frac{\hbar\Omega}{2}\left(\hat{\sigma}_+ \otimes \hat{a} + \hat{\sigma}_- \otimes \hat{a}^\dagger\right) \\ &= \frac{\hbar\Omega}{2}\begin{pmatrix} 0 & \hat{a} \\ \hat{a}^\dagger & 0 \end{pmatrix} \end{aligned} \tag{59}$$

where Ω is the Rabi frequency and $\hat{\sigma}_\pm$ are the atomic raising and lowering operators.

In order to combine the Hilbert spaces for these two separate quantizes systems, block-operator notation is applied. Notice that $\hat{H}_{Atom}$ acts on the two-state atomic basis, while $\hat{H}_{Photon}$ acts on the infinite Fock basis of photons. We then must combine the two bases in $\hat{H}_{Interaction}$ which describes the creation of a photon in the Fock basis upon atomic emission, and the annihilation of a photon upon absorption. So this Hamiltonian should not be thought of as a two-by-two matrix acting on the atomic basis, but rather an infinite matrix enumerating the interactions between all possible states. All together, the final Hamiltonian is then [**?**]

$$\begin{aligned} \hat{H}_{JC} &= \frac{\hbar\omega}{2}\hat{\sigma}_z + \hbar\omega\hat{a}^\dagger\hat{a} + \frac{\hbar\Omega}{2}\left(\hat{\sigma}_+ \otimes \hat{a} + \hat{\sigma}_- \otimes \hat{a}^\dagger\right) \\ &= \frac{\hbar}{2}\begin{pmatrix} \omega & \Omega\hat{a} \\ \Omega\hat{a}^\dagger & -\omega \end{pmatrix} + \hbar\omega\hat{a}^\dagger\hat{a} \end{aligned} \tag{60}$$

This is not the whole story, however. Any real cavity has some photonic decay rate, meaning we must include dissipative operators and apply Lindbladian dynamics to solve for a steady state via the master equation. Our dissipative operator then becomes $L = \sqrt{\gamma}\hat{a}$ where γ gives the decay rate such that our master equation is [**?**]

$$\dot{\rho} = -i\left[\hat{H}_{JC}, \rho\right] + \gamma\left(\hat{a}\rho\hat{a}^\dagger - \frac{1}{2}\hat{a}^\dagger\hat{a}\rho - \frac{1}{2}\rho\hat{a}^\dagger\hat{a}\right). \tag{61}$$

To learn the QuTiP package, the Jaynes-Cummings model was coded (see Code Block. **??**) based on an implementation by Robert Johansson [**?**]. This code differs slightly from the above description in that the atomic Hamiltonian basis is shifted such that it is in the rotating frame $\hat{H}_{Atom} = \hbar\omega|e\rangle\langle e|$.

Code Block 1: Python code for the Jaynes-Cummings Model

```
# Initial Arguments
w_a = 2.0
w_c = 2.0
W = 0.01
gamma = 0.005
xi = 0.0025
eta = 0.05
h = 1.0
N = 10 # Truncate the Fock basis to N photons
times = np.linspace(0, 2500, 10000)

# Set up operators in the relevant basis
a = tensor(qeye(2), destroy(N)) # Acts on photon Fock basis with N states as
    lowering operator
s_minus = tensor(destroy(2), qeye(N)) # Acts on atomic basis with 2 states as
    lowering operator
s_atom = tensor(Qobj([[0, 0], [0, 1]]), qeye(N)) # Acts on atomic basis with a
    shifted ground states

# Set up the Hamiltonian
H = (h * w_a) * s_atom + (h * w_c) * a.dag() * a + (h * W) * (s_minus.dag() * a
   + s_minus * a.dag())

# Set up dissipative operator for a lossy cavity
cavity_decay = np.sqrt(gamma) * a
c_ops = [cavity_decay]

# Set up the expectation value operators
number_operator = a.dag() * a
atomic_state_operator = s_minus.dag() * s_minus
e_ops = [number_operator, atomic_state_operator]

# Start the system in an initially excited state with 0 photons
psi_0 = tensor(basis(2,1), basis(N, 0))

# Time evolve the system using qutip mesolve
result = mesolve(H, psi_0, times, c_ops=c_ops, e_ops=e_ops)
photons = result.expect[0]
```

This code uses the `mesolve` function for time evolving a quantum system such that we may inspect the evolution of the state. For an initial state where the atom is excited and there are no photons present in the cavity, we obtain decaying Rabi oscillations as expected (see Fig. **??**).

This code will be a base for all quantum code going forward, but before we can dive any deeper into quantized systems, we must take a step back to look at a semi-classical system.

B. Electromagnetically Induced Transparency in a Λ-system with Classical Driving Fields

Consider a quantized 3-level Λ-system of states driven by strong external electromagnetic fields. We consider the fields to be so strong that the quantum mechanical nature of the beams is reduced to classical behaviour in accordance with Ehrenfest's theorem (i.e. the correspondence principle). This makes the system semi-classical since the atomic system is quantized while the driving fields are treated classically.

The Hamiltonian for this system in the atomic basis can be obtained by writing down the diagonal atomic Hamiltonian and shifting it such that $\langle 1|\hat{H}|1\rangle = 0$, then adding on the effects of the electromagnetic fields which drive the allowed transitions $|1\rangle \leftrightarrow |3\rangle$ and $|2\rangle \leftrightarrow |3\rangle$ at frequencies $\omega_s \approx \omega_{13}$ and $\omega_c \approx \omega_{23}$. The rotating wave approximation may then be applied to simplify the off-diagonal transitional entries as Rabi frequencies Ω_s, Ω_c, and a unitary transformation may be applied to write the diagonal entries in terms of the detunings due to the driving beams $\Delta_s = \omega_{13} - \omega_s$ and $Delta_c = \omega_{23} - \omega_c$. A similar procedure is applied to obtain the Hamiltonian for the 4-level system of focus in

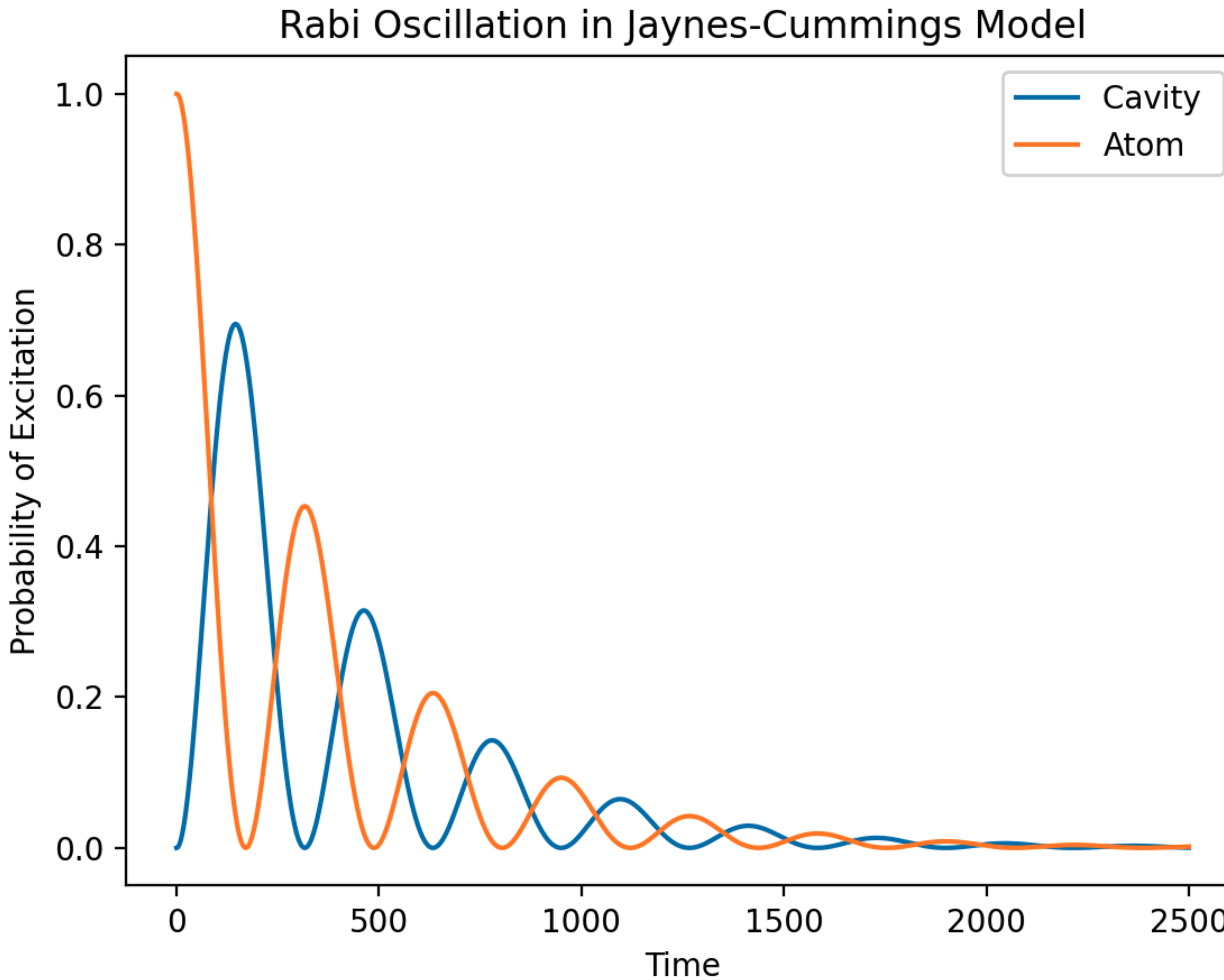

FIG. 7: Decaying Rabi oscillation produced by the Jaynes-Cummings model with initial state $|\psi\rangle_0 = |e\rangle \otimes |0\rangle$.

the appendix **??**.

The Hamiltonian in this form is well known to be [**?**]

$$\hat{H} = -\frac{\hbar}{2} \begin{pmatrix} 0 & 0 & \Omega_s \\ 0 & -(\Delta_s - \Delta_c) & \Omega_c \\ \Omega_s^* & \Omega_c^* & \Delta_s \end{pmatrix}. \tag{62}$$

In order to consider the effect of the environment for Lindbladian time evolution, we must mathematically express the decay operators of the system. We have decay operator Γ_{13} which acts on transition σ_{13} and Γ_{23} which acts on σ_{23}. We can then follow the template given by the Jaynes-Cummings model, changing out the `mesolve` time evolution function for the `steadystate` function which numerically solves the Lindblad master equation.

This results in a density matrix, which can then be used to obtain the expectation value of the dipole transition operator $\hat{d}_{13} \propto \sigma_{13}$. The linear electric susceptibility $\chi^{(1)}$ of the system is proportional to the square of the expectation value of the dipole transition operator $\hat{d}_{13}$, which itself is proportional to the expectation value $\langle\sigma_{13}\rangle$ [**?**].

By setting Δ_c to be fixed, we can scan over Δ_s values to find the optimal conditions for EIT at the given Δ_c (and any other necessary parameters). Solving for the steady state then expectation value at each detuning produces fig. **??**.

TODO - Talk about figure and relation to EIT

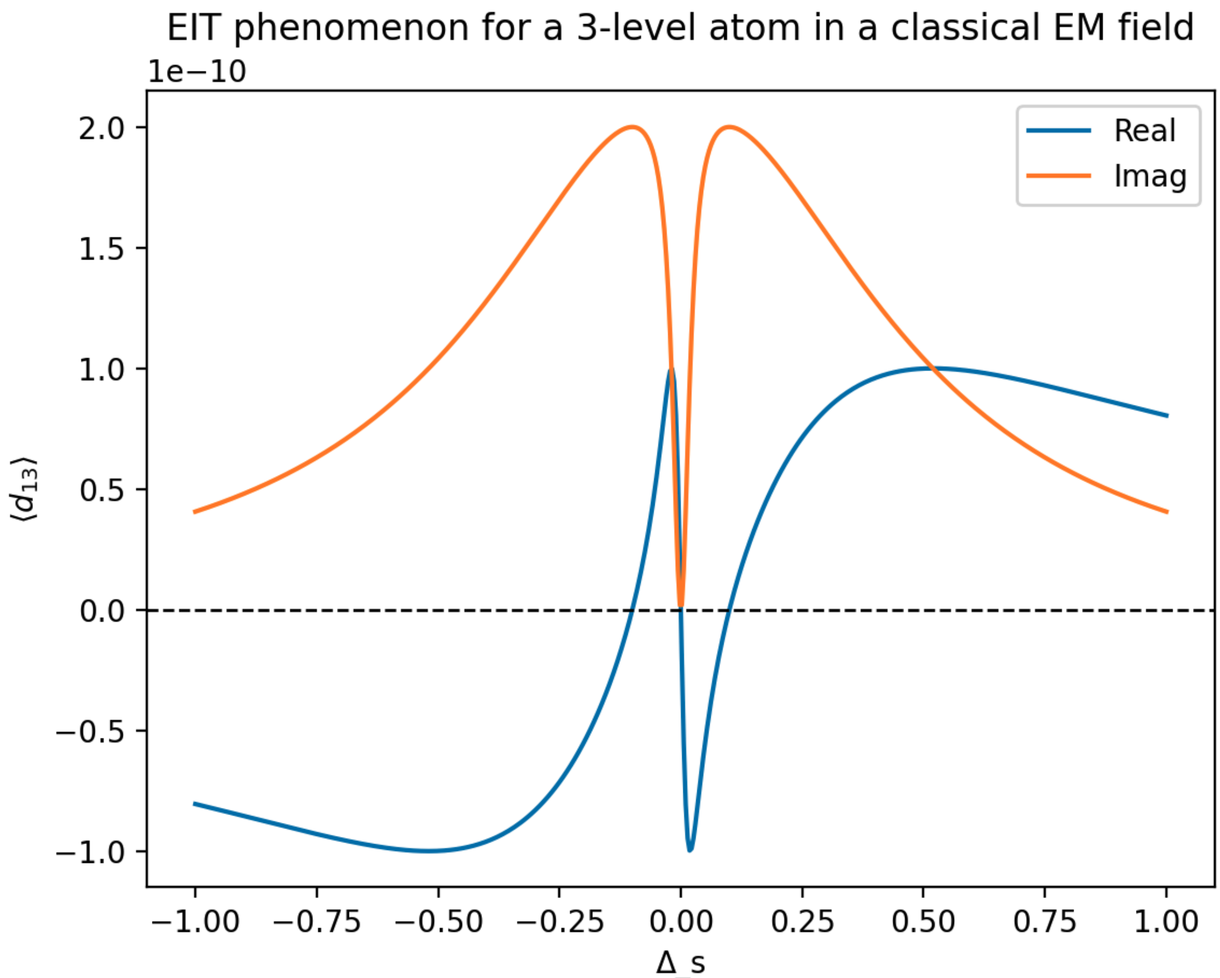

FIG. 8: TODO - Explain figure

C. Cavity Electromagnetically Induced Transparency in a Λ-system with Quantized Driving Fields

TODO - Write what we did here to reproduce the results

D. IDK

We will treat our atom as a three-level system with discrete states while lights behaves as a continuous field such that we do not need to discretize light in the Fock basis of photon number.

The polarization vector of the atomic cloud may be given by

$$\vec{P} = N\langle\hat{\vec{d}}\rangle = \epsilon_0\chi\vec{E} \tag{63}$$

where N is atomic density, $\hat{\vec{d}}$ is the average dipole moment, and χ is the magnetic susceptibility [?]. One can find that for a three-level atom (TODO derive or cite?)

$$\chi = \frac{-Nd^2}{\epsilon_0\hbar}\frac{\delta}{|\Omega_c|^2 - \delta(\Delta + i\Gamma)} \tag{64}$$

where TODO - Define detunings and dissipation rate along with a figure for a three-level atom.

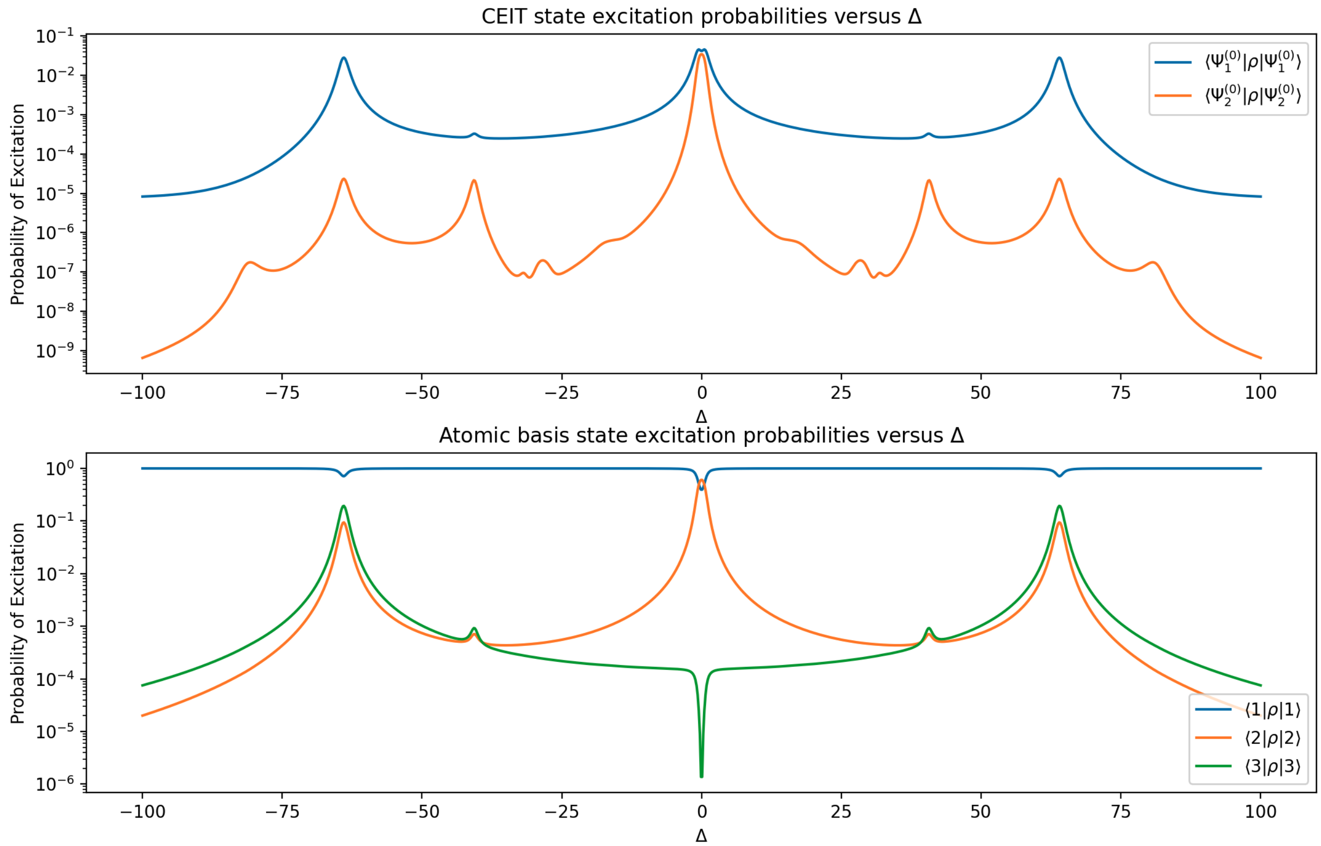

FIG. 9: TODO

The Hamiltonian of the three-level atom can be given as

$$H_{atom} = TODO(diagonal - slides) \tag{65}$$

and the Hamiltonian of the light-matter interaction can be given as

$$H_{atom} = TODO(see_slides) \tag{66}$$

The dissipative operators...

Solve master equation with...

Here's the plot...

E. Meeting 2 with MacRae

1. *Hamiltonian derivation with light as a classical field*

To construct the Hamiltonian, it is advantageous to first express it as a sum of the contributions to be considered

$$\hat{H} = \hat{H}_{Atom} + \hat{H}_{Light\ Atom\ Interaction} + \hat{H}_{Light}. \tag{67}$$

Since we are considering the simplified system where light acts as a continuous field, all we care about is how that field interacts with the atom. So at this point, we choose to neglect the term $\hat{H}_{Light}$ to obtain

$$\hat{H} = \hat{H}_{Atom} + \hat{H}_{Light\ Atom\ Interaction}. \tag{68}$$

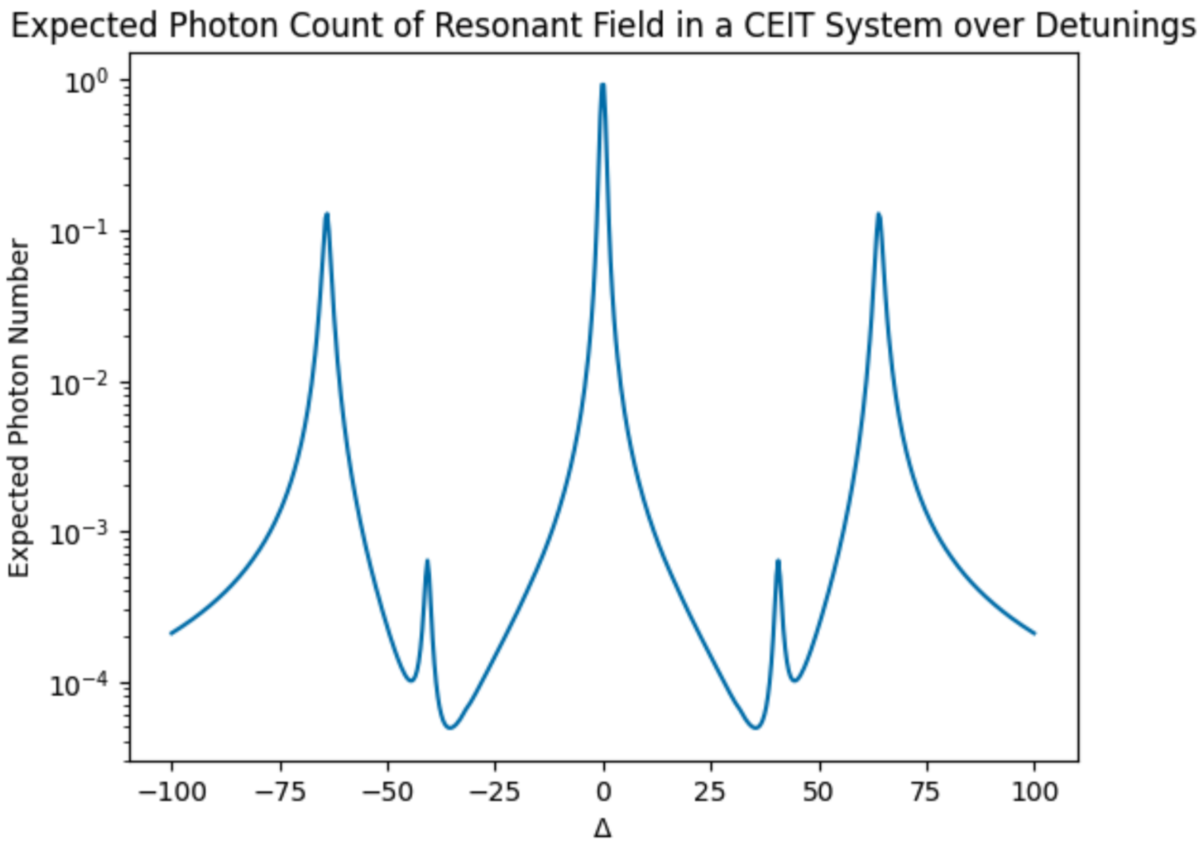

FIG. 10: TODO

Let $\hat{\sigma}_{i,j} = |i\rangle\langle j|$ express the atomic transition operators which operate on state vectors to transition from state j to i. For example,

$$\hat{\sigma}_{1,3} = |1\rangle\langle 3| = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{69}$$

and acting this operator on basis state $|i\rangle$ yields

$$\hat{\sigma}_{1,3}|i\rangle = |1\rangle\langle 3 \mid i\rangle = \delta_{3,i}|1\rangle \tag{70}$$

where δ denotes the usual Kronecker delta.

TODO

3-level system frequency 2-3 $= \Omega_c$. $\Gamma_{3->2}$ and $\Gamma_{3->1}$ emission. $\hat{\sigma}_{i,j} = |i><j|$. $\Delta\omega_s - \omega_{3,1}$. $\delta = \Delta_c - \Delta$

$\hat{H} = \hat{H}_a + \hat{H}_{LightAtom} + \hat{H}_{Light}$

Neglect $\hat{H}_{Light}$

$\hat{H}_a = \hbar(\omega_{3,1} - \omega_{3,1})\hat{\sigma}_{2,2} + \hbar\omega_{3,1}\hat{\sigma}_{3,3}$

$\hat{H}_{LightAtom} = -\hat{\vec{d}}_{1,3} \cdot \vec{E}_s - \hat{\vec{d}}_{2,3} \cdot \vec{E}_c$

Expand dipole operators and E in terms of complex exponentials.

$\hbar\Omega_s = \vec{d}_{1,3} \cdot \vec{E}_s$

$\hbar\Omega_c = \vec{d}_{2,3} \cdot \vec{E}_c$

Plug in Ω_i values into hamiltonian, replacing dipoles and E

We need the interacting picture

$\hat{H} = \hat{H}_0 + \hat{H}_I$

$\hat{H}_0 = \omega_s\hat{\sigma}_{3,3} + (\omega_s - \omega_c)\hat{\sigma}_{2,2}$

$\frac{\hat{H}_I}{\hbar} = \frac{\hat{H}-\hat{H}_0}{\hbar} = (\omega_{3,1} - \omega_s - \omega_s + \omega_{3,2})\hat{\sigma}_{2,2} + (\omega_{3,1} - \omega_s)\hat{\sigma}_{3,3} + \hat{H}_{LightAtom}$
Plug in Δ and δ then $\hat{H}_{LightAtom}$ to obtain interaction hamiltonian
$\hat{H}_I = \begin{pmatrix} 0 & 0 & \Omega_s e^{i\omega_s t}\tilde{\Omega}_s e^{i\omega_s t} \\ 0 & \delta & \Omega_c e^{i\omega_c t}\tilde{\Omega}_c e^{i\omega_c t} \\ c.c & c.c & \Delta \end{pmatrix}$
Then
$V = e^{-i\hat{H}_0 t}\hat{H}_I e^{i\hat{H}_0 t}$
Notice that $e^{-i\hat{H}_0 t}$ is diagonal so this is simple
Multiply through, then apply rotating wave approximation to obtain
$\hat{V} = \begin{pmatrix} 0 & 0 & \Omega_s \\ 0 & \delta & \Omega_c \\ c.c & c.c & \Delta \end{pmatrix}$

2. a

Density matrix times an operator
$\hat{\rho}\hat{A} = |\psi > (< \psi|\hat{A})$
$\hat{B} = I\hat{B} = \sum_c |c > (\hat{B}|c >)^\dagger$
Do some math, then we find that
$< \hat{A} >= Tr[\hat{\rho}\hat{A}]$
For linear media, the polarization is $\vec{D} = \epsilon_0\vec{E} + \vec{P} = \epsilon_0(1+\xi)\vec{E} = \epsilon\vec{E}$ and $\vec{P} = N < \hat{\vec{d}} >$
$1 + \xi = \frac{\epsilon}{\epsilon_0}$
For non-magnetic materials $\mu \approx \mu_0$, so
$1 + \xi = \frac{\epsilon}{\epsilon_0} = \frac{\epsilon\mu}{\epsilon_0\mu_0} = \left(\frac{c}{v}\right)^2 = n^2$
Something like this, check all these derivations
Write up derivation to the point that we obtain
$\xi = \frac{N<d>}{\epsilon_0\hbar\Omega}$
$< \vec{d} >= Tr[\hat{\rho}(\hat{\vec{d}}\hat{\sigma}_{i,j} + h.c.)]$
Expand out (Find a textbook with this) and we obtain
$\xi = \frac{Nd^2}{\epsilon_0\hbar\Omega}\rho_{i,j}$
The density matrix should be a unitary

F. Meeting 3 with MacRae

Note that Polarization is real as $P = P_0 e^i wt + P_0^* e^- iwt$ as real valued

We use the result we obtained $< d >$ with Beer's Law to obtain $E(t)$ (wave propagation, phase shift, and absorption). We get no phase shift and no absorption for perfect EIT. In reality, we send a pulse through. Some of the pulse frequencies will be absorbed! The longer the pulse, the less we lose. This is the effect of the medium responding.

For a pulse of length $\Delta\tau$. If $\Delta\tau = 10^{-5}$ then the light pulse will be about 3 km long! We need the pulse to fit in the EIT window. Slowed light helps here such that the pulse fits in the cell. Widening the frequency window increases the speed of the slowed light, so we need a very precise frequency. The narrower the transparency window, the greater the group velocity (see the equation for group velocity dependent on dn/dw).

We can however include many more atoms to increase n to lower group velocity. This is called 'increasing optical depth'.

$$v_g = \frac{c}{n + \omega\frac{dn}{d\omega}} \tag{71}$$

V. PROBLEM DEFINITION

TODO - Write this

TODO - It may be the case that the system is Doppler insensitive as is the case with Λ-systems. Do the math to find out if this may be the case!!

VI. TEMP SECTION - NEXT STEPS

I have covered a sufficient amount of background and written enough on it. My meeting notes document to sufficient detail that I can write them out formally to approximately the same degree now vs later. If anything, they will benefit from being written later with hindsight into how things actually work.

So, I can leave the thesis here for now. My next steps then are the following:

- Read and re-read relevant CEIT papers
- Find how to model the three-level system within a cavity using classical electromagnetic fields with dissipative operators for Lindbladian time evolution
- Find how to quantize the system (optional - but probably in the papers I will read)
- Read FWM papers
- Learn how to computationally model FWM

VII. SEMI-CLASSICAL THREE-LEVEL ATOM

We can begin to understand the behaviour of EIT in three-level Λ-type atoms through semi-classical modelling. We can consider a single three-level atom with quantized states in the presence of a classically oscillating electromagnetic field

$$\vec{E} = \vec{E}_0 e^{i\omega t}. \tag{72}$$

In this limit, we consider E_0 to be sufficiently large such that photon quantization can be neglected (i.e. in the classical limit, the behaviour of the quantized electromagnetic field must converge to the classical behaviour). We may express the full Hamiltonian as the sum

$$\hat{H} = \hat{H}_{Atom} + \hat{H}_{Light} + \hat{H}_{Light\ Atom\ Interaction}. \tag{73}$$

We may the neglect the term $\hat{H}_{Light}$ to approximate our full Hamiltonian since the state of the electromagnetic field is not significantly affected by the atom in the (i.e. there are is little coupling between the state of the laser field and the single atom). Our Hamiltonian then becomes

$$\hat{H} = \hat{H}_{Atom} + \hat{H}_{Light\ Atom\ Interaction}. \tag{74}$$

where TODO - Define ω values?

$$\begin{aligned}\hat{H}_{atom} &= \omega_s \hat{\sigma}_{3,3} + (\omega_s - \omega_c)\hat{\sigma}_{2,2} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & \omega_s - \omega_c & 0 \\ 0 & 0 & \omega_s \end{pmatrix}\end{aligned} \tag{75}$$

and

$$\hat{H}_{Light\ Atom\ Interaction} = -\hat{\vec{d}}_{1,3} \cdot \vec{E}_s - \hat{\vec{d}}_{2,3} \cdot \vec{E}_c \tag{76}$$

TODO - Finish rest of derivation!!

Thus, we obtain

$$\hat{H} = \begin{pmatrix} 0 & 0 & \Omega_s \\ 0 & \delta & \Omega_c \\ \Omega_s^* & \Omega_c^* & \Delta \end{pmatrix} \tag{77}$$

TODO - This has been coded, include code snippets here and talk about results.

A. Cavity EIT with a Λ-System

TODO - change line numbers relevant to final code TODO - Place in appendix

Code Block 2: Python code for CEIT with a three-level atom

```
########
# This implementation is based off of Coherent Control of Quantum Fluctuations
   Using Cavity Electromagnetically Induced Transparency
   (https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.111.113602)

# Initial Arguments
kappa = 1.0 # Write all parameters in terms of kappa
W_c = 40.0 * kappa
g = 50.0 * kappa
Delta = 0.01
epsilon = 1.0 * kappa
Gamma_31 = 0.1 * kappa
Gamma_32 = 0.1 * kappa
N = 7 # Truncate the Fock basis to N photons

# Atomic basis kets
atomicKet1 = basis(3, 0) # |1>
atomicKet2 = basis(3, 1) # |2>
atomicKet3 = basis(3, 2) # |3>

# Set up operators in the relevant basis
s11 = tensor(atomicKet1 * atomicKet1.dag(), qeye(N))
s22 = tensor(atomicKet2 * atomicKet2.dag(), qeye(N))
s33 = tensor(atomicKet3 * atomicKet3.dag(), qeye(N))
s13 = tensor(atomicKet1 * atomicKet3.dag(), qeye(N)) # Photon absorption 1->3
s23 = tensor(atomicKet2 * atomicKet3.dag(), qeye(N)) # Photon absorption 2->3
s31 = s13.dag() # Photon emission 3->1
s32 = s23.dag() # Photon emission 3->2
a = tensor(qeye(3), destroy(N)) # Acts on photon Fock basis as lowering operator

# Decay operators
cavity_decay = np.sqrt(2*kappa) * a
atomic_decay_13 = np.sqrt(2*Gamma_31) * s13
atomic_decay_23 = np.sqrt(2*Gamma_32) * s23
c_ops = [cavity_decay, atomic_decay_13, atomic_decay_23]

def eta(n):
    return g * np.sqrt(n/2) / W_c

def steady_evo(num_vals):
    state_1_vals = []
    state_2_vals = []
    state_3_vals = []
    state_4_vals = []
    state_5_vals = []
    real_dipole = []
    imag_dipole = []
    photon_numbers = []
    Delta_vals = np.linspace(-100.0, 100.0, num_vals)

    # Useful basis vectors
    k_s1_n0 = tensor(atomicKet1, basis(N, 0))
    k_s2_n0 = tensor(atomicKet2, basis(N, 0))
```

```
    k_s3_n0 = tensor(atomicKet3, basis(N, 0))
    k_s2_n1 = tensor(atomicKet2, basis(N, 1))
    k_s3_n1 = tensor(atomicKet3, basis(N, 1))

    for Delta in Delta_vals:
        # Get system for Delta value
        # Note that we assume perfect cavity resonance (Delta_1, Delta_2 = 0)
        H = -Delta * s11 + \
            Delta * a.dag() * a + \
            epsilon * (a + a.dag()) + \
            g * (a * s31 + a.dag() * s13) + \
            W_c * (s32 + s23)

        # Compute steady state solution
        rho = steadystate(H, c_ops, method='direct')

        # Find phase for n=0 and n=1
        expect_n0 = expect(k_s2_n0 * k_s3_n0.dag(), rho)
        phi_n0 = np.angle(expect_n0)
        expect_n1 = expect(k_s2_n1 * k_s3_n1.dag(), rho)
        phi_n1 = np.angle(expect_n1)

        # THIS IS GOOD
        num_op = tensor(qeye(3), num(N))
        photon_numbers.append(expect(num_op, rho))

        # TODO - This does not seem quite right - MacRae said real and imag
           look swapped
        # dipole = expect(k_s1_n0 * k_s3_n0.dag(), rho)
        dipole = expect(k_s3_n0 * k_s1_n0.dag(), rho)
        real_dipole.append(dipole.real)
        imag_dipole.append(dipole.imag)

        # Define rotated plus minus states in full Hilbert space
        plus_n0 =  (k_s2_n0 + np.exp(-1j*phi_n0)*k_s3_n0) / np.sqrt(2)
        minus_n0 = (k_s2_n0 - np.exp(-1j*phi_n0)*k_s3_n0) / np.sqrt(2)
        plus_n1 =  (k_s2_n1 + np.exp(-1j*phi_n1)*k_s3_n1) / np.sqrt(2)
        minus_n1 = (k_s2_n1 - np.exp(-1j*phi_n1)*k_s3_n1) / np.sqrt(2)

        # psi^(0)_1
        psi_1 = tensor(atomicKet1, basis(N, 1)) - \
                eta(1) * plus_n0 + \
                minus_n0

        # psi^(0)_2
        psi_2 = tensor(atomicKet1, basis(N, 2)) - \
                eta(2) * plus_n1 + \
                minus_n1

        # Normalization
        psi_1 = psi_1.unit()
```

VIII. MEETING DEC 3

Classical susceptibility / index of refraction within cavity calculation for sanity check. We can calculate quantum stuff, but we need something to check against. This is the purpose of this step.

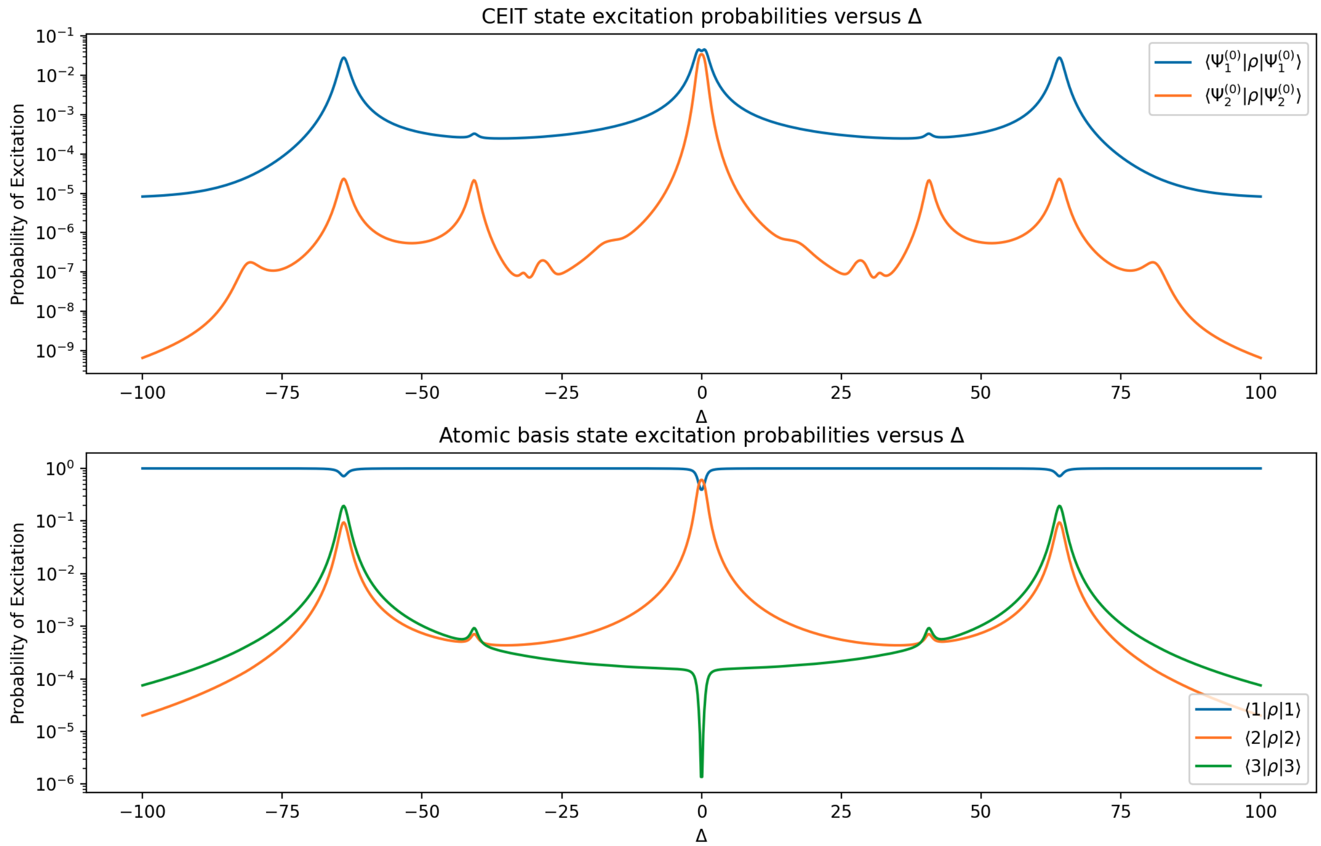

FIG. 11: TODO

What kind of imperfections will kill the result? When does decoherence occur in classical model?

We might want to find photon number in quantum case to find light intensity.

Plotting Fig 1 with number operators matches Fig 1 peaks in paper. This is what we want

Next step: Get classical result that we can compare against later for sanity check (they should match). This also tells us if the experiment is possible in the first place. If parameters don't work out, then it may not be possible.

IX. MEETING JAN 8

With 2 driving beams we cannot use RWA

$k = n(\omega)\frac{\omega}{c}$ in a medium

We need to conserve momentum and energy (frequency) to get phase matching

If we have rubidium atoms in an optical resonator, can we predict the squeezing?

Go through papers

X. MEETING JAN 12

The goal is now conversion efficiency. Look into Capckuck's stuff. Squeezing is gravy (if we can get it). Effieceincy calculations of a upconversion -> fiber optic transmission -> downconversion (end-to-end conversion efficiency) would be awesome!

XI. MAIN RESULTS FROM RESEARCH BLOCK

I read a handful of papers and here are some main points that stuck out.

Paper list:

- Microscopic Theory of Squeezed Light in Quantum Dot Systems
- Theory of cavity-enhanced spontaneous four wave mixing
- Generation of 87Rb resonant bright two-mode squeezed light with four-wave mixing
- Quantum Noise and Correlations in Resonantly Enhanced Wave Mixing Based on Atomic Coherence
- Slow light propagation and amplification via electromagnetically induced transparency and four-wave mixing in an optically dense atomic vapor
- (Didn't read fully, but used) Parametric amplification in an electromagnetically-induced-transparency medium
- (Didn't read fully, but used) Competition between electromagnetically induced transparency and Raman processes

A. Microscopic Theory of Squeezed Light in Quantum Dot Systems

They give operators for quadratures which need to be squeezed. MacRae said that these are conventional and we would likely use the same ones $\hat{X} = \frac{1}{2}(a + a^\dagger)$ and $\hat{Y} = \frac{1}{2i}(a - a^\dagger)$. The amount of squeezing is then expressed by getting expressions for ΔX and ΔY, which have difference $\Delta X - \Delta Y = 4Re(\delta\langle aa\rangle)$ where $\delta\langle aa\rangle = \langle aa\rangle - \langle a\rangle\langle a\rangle$. They also give the operator for the $g^{(2)}(0)$ correlations (simultaneous photon emission operator)

$$g^{(2)}(0) = \frac{\langle a^\dagger a^\dagger aa\rangle}{\langle a^\dagger a\rangle\langle a^\dagger a\rangle}$$

They also give a Hamiltonian for their system (could be useful).

Their data shows that different system parameters can result in different amounts of squeezing. This tells me that we need to balance parameters to obtain significant squeezing, FWM, and EIT all at once.

THey make a statement that MacRae thought was useful: 'In practice, amplitude squeezing is maintained only if classical pump relative-intensity noise (RIN) is suppressed below shot noise.' He said this may be difficult, but should be possible.

Another result is stated as 'This underscores the need for narrow-linewidth sources and good thermo-mechanical stability to preserve quadrature squeezing in the presence of injection-induced phase noise.' Also 'From an engineering perspective, three levers govern performance, cavity decay, seed strength, and dephasing.'

B. Theory of cavity-enhanced spontaneous four wave mixing

This paper focuses on a system which I think is quite different from what we will be working with. Some mathematical techniques they used (Fourier transform techniques and interpretations) may be useful to us.

One result is 'Thus, for a higher coefficient of finesse, there will be more cavity round trip times, and the dimension of the Hilbert space which describes each photon will increase.' which seems straightforward to me. They then go on to say 'Thus, interestingly, this physical system leads to higher-dimensional entanglement in the temporal-mode degree of freedom with a dimension which can be tailored by adjusting the cavity finesse.'

Stated another way 'increasing the finesse leads to more signal and idler cavity round-trips and hence to a greater correlation time.'

They calculate the flux out of the cavity based on a few parameters and give some different regimes. Could be useful.

C. Generation of 87Rb resonant bright two-mode squeezed light with four-wave mixing

This paper focuses on using a double-Λ-system to produce FWM far off resonance in 85Rb such that the output photons produced are resonant with a transition of an 87Rb double-Λ-system which is more useful for quantum memory. They made a statement that I thought was very important: 'While the original proposals for the double-lambda configuration [41–43] suggest that it should be possible to use EIT to eliminate absorption when operating on resonance, in practice this is not the case and residualabsorption dominates over the FWM process whenthe probe frequency is tuned closer to resonance.'

So avoid needing to use EIT, they tune far off resonance to prevent significant absorption in the 85Rb sample material but in a regime where they can still obtain FWM and squeezing. They then use EIT with the 87Rb for quantum memory.

Not in this paper, but related to quantum memory: MacRae said that we can create quantum memory with 87Rb and EIT by having a signal and control beam which put the system into a dark state, then if we send in a pulse of light with quantum information encoded, then turn off the control beam, we can get short-lived quantum memory storage. Once the control beam is turned back on, the pulse is re-emitted from the material. There is some decoherence of course, making this storage method somewhat short-lived, but it could be useful in quantum optical circuits.

They later state that there are multiple competing non-linear processes (including FWM) which makes it difficult to get FWM alone.

D. Quantum Noise and Correlations in Resonantly Enhanced Wave Mixing Based on Atomic Coherence

This paper looks into supressing quantum noise fluctuations via non-linear optical processes. Squeezing can be used for limited noise reduction. Quantum coherence and interference can be used to improve the degree of noise reduction. They generate Stokes and anti-Stokes fields with a near complete supression of quantum fluctuations and a perfect correlation structure (100% squeezing).

They use a double-Λ-system with FWM to produce squeezing. They use quite a bit of math, make some approximations, and show a regime where those approximations hold to obtain the desired correlation structure. They state that their methods can be applied in the few-photon limit, meaning that the theory may allow for single-photon quantum contol via non-linear optical processes.

E. Slow light propagation and amplification via electromagnetically induced transparency and four-wave mixing in an optically dense atomic vapor

They state 'In the limit of low optical depth, it is sufficient to take into account only the effects of this single Λ. However, many applications require operation at high optical depth [9, 10], where additional nonlinear effects may become important [11, 12, 13, 14]. One such effect is resonant fourwave mixing a nonlinear process arising from the far off-resonant interaction of the control field.' so they use a double-Λ-system

They also state 'under certain conditions, FWM may lead to gain for both the signal and Stokes fields, which could compensate for any optical losses'

They use some math to solve, including Floquet theory. They express a Hamiltonian for their system (may be helpful). They express as a system of differential equations and solve for analytic expressions to fully describe the propagation of light through the medium. They then give conditions for EIT and FWM based on the equations they obtained.

They importantly state 'Spectra taken at higher optical depth reveal more clear evidence of the constructive and destructive interference between EIT and FWM.' EIT and FWM can compete if the Raman paths destructively interfere (is what I gather from this).

They compare theory against expreimental results, which match quite well, but indicate that they may not be considering some processes for some cases. They state that signal field absorption is underrepresented by the model, and that they may not be considering some decay mechanism.

They state 'modeled by a simple double-Λ system, where the output signal and stokes field amplitudes are the results of interference of "traditional" EIT and FWM.'

F. Parametric amplification in an electromagnetically-induced-transparency medium

They state: 'Experimental and theoretical analyses have revealed that, while the EIT picture is valid for relatively low atomic densities, the PA picture becomes dominant for high atomic densities and high coupling powers. We have also found that the parametric coupling between the probe and Stokes waves can be nullified by the proper choice of the coupling detuning frequency, due to destructive interference of two elementary processes.' and 'The last terms of the two equations represent the parametric coupling, and if we neglect them, Eq. 2 is simply reduced to the ordinary EIT process, i.e., the linear absorption of each wave is cancelled by the presence of the factor c. In the limit of c →1, the medium becomes transparent for both the probe and Stokes waves, but it never amplifies them. If the last terms are included, however, the situation is totally different. In this case the two waves are parametrically coupled and the two waves can be amplified by the proper choice of parameters.'

G. Competition between electromagnetically induced transparency and Raman processes

They state 'Often, the various processes compete with each other, whereby some processes are suppressed or interference between processes renders the medium transparent to the applied fields. Well-known examples include competition between third-harmonic generation and multiphoton ionization,and between four-wave mixing and two-photon absorption. Very recently, Harada et al. demonstrated experimentally that stimulated Raman scattering can disrupt electromagnetically induced transparency EIT, where the incident probe beam is depleted and new fields are generated via Raman processes.'

XII. HAMILTONIAN FORMULATION

XIII. COMPUTATIONAL METHODS

XIV. RESULTS

XV. DISCUSSION

A. Conditions for Optical Squeezing

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Appendix A: Hamiltonian Formulation

TODO - I MESSED UP THE STATE DIAGRAM!! WE WANT TO USE THE 795 nm TRANSITION FOR SIGNAL AND THE 1530 FOR IDLER!!! CHANGE FOR THIS!!!

The first step in Hamiltonian formulation is system definition. We are modelling a 4-level diamond system as depicted in Fig. ??. For this system, the choice of the ground state energy is arbitrary, and all other energies can be defined in terms of their differences as $\omega_2 = \omega_1 + \omega_{12}$, $\omega_3 = \omega_1 + \omega_{13}$, and $\omega_4 = \omega_2 + \omega_{24} = \omega_3 + \omega_{34}$.

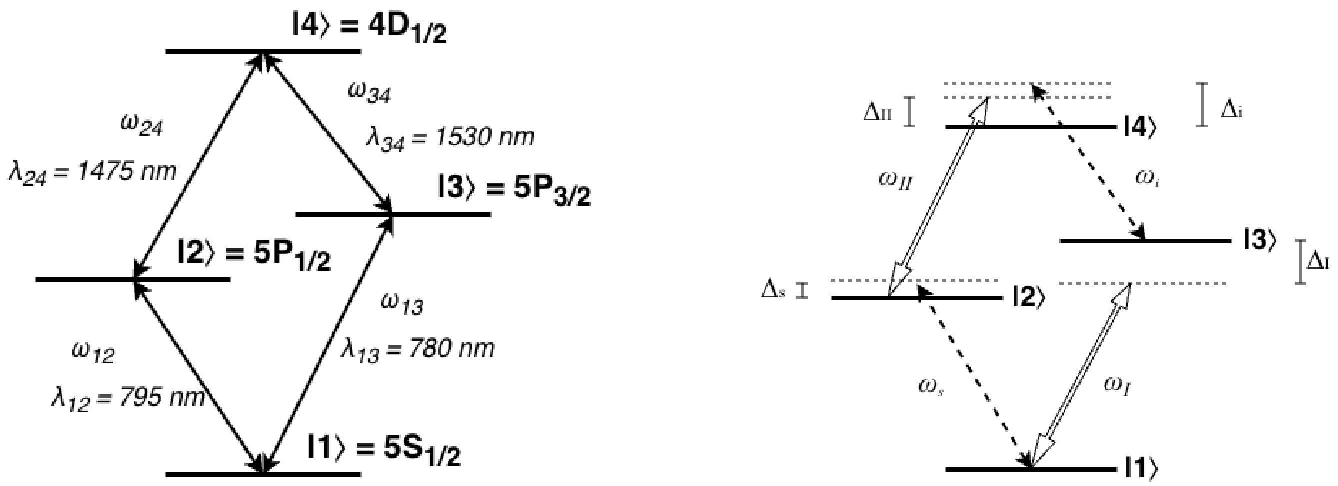

FIG. 12: Diamond configuration atom diagrams. (Left) Duplicate of Fig. ?? placed for convenience. A diamond configuration atomic system with non-degenerate energy levels $|1\rangle, |2\rangle, |3\rangle, |4\rangle$. The system has allowed transitions $|1\rangle \leftrightarrow |2\rangle$, $|1\rangle \leftrightarrow |3\rangle$, $|2\rangle \leftrightarrow |4\rangle$, and $|3\rangle \leftrightarrow |4\rangle$ with respective energies $\hbar\omega_{12}$, $\hbar\omega_{13}$, $\hbar\omega_{24}$, and $\hbar\omega_{34}$ and corresponding wavelengths λ_{12}, λ_{13}, λ_{24}, and λ_{34}. (Right) A strong pump drives transition $|1\rangle \leftrightarrow |3\rangle$ at frequency ω_I with detuning $\Delta_I = \omega_I - \omega_{13}$, a separate strong pump drives transition $|2\rangle \leftrightarrow |4\rangle$ with detuning $\Delta_{II} = \omega_{II} - \omega_{24}$, a weak signal beam drives transition $|1\rangle \leftrightarrow |2\rangle$ with detuning $\Delta_s = \omega_s - \omega_{12}$, and a weak idler field (undriven) is induced by the presence of the other fields through a FWM process, resulting in transition $|3\rangle \leftrightarrow |4\rangle$ with emission detuning $\Delta_i = \omega_i - \omega_{34}$.

These states can be described without any interactions by the simple Hamiltonian

$$\hat{H}_0 = \hbar\omega_1\hat{\sigma}_{11} + \hbar\omega_2\hat{\sigma}_{22} + \hbar\omega_3\hat{\sigma}_{33} + \hbar\omega_4\hat{\sigma}_{44} \tag{A1}$$

$$= \hbar\begin{pmatrix} \omega_1 & 0 & 0 & 0 \\ 0 & \omega_2 & 0 & 0 \\ 0 & 0 & \omega_3 & 0 \\ 0 & 0 & 0 & \omega_4 \end{pmatrix} \tag{A2}$$

The reason we are attempting to obtain optical squeezing in the up-converted idler field is to transmit quantum information at telecom wavelengths. Therefore, we care about the quantized fields of the signal (input carrier of quantum information) and idler (output of frequency-converted quantum information) beams. Apply the theoretical resonant cavity approximation such that we consider photons in the Fock bases of the exact frequencies of the signal and idler beams to obtain the Hamiltonian

$$\hat{H}_{photon} = \hbar\omega_{13}\hat{a}^\dagger\hat{a} + \hbar\omega_{24}\hat{b}^\dagger\hat{b} \tag{A3}$$

We then have the Hamiltonian describing these separate systems as

$$\hat{H}_S = \hat{H}_0 + \hat{H}_{photon} \tag{A4}$$

$$= \hbar\omega_1\hat{\sigma}_{11} + \hbar\omega_2\hat{\sigma}_{22} + \hbar\omega_3\hat{\sigma}_{33} + \hbar\omega_4\hat{\sigma}_{44} + \hbar\omega_{13}\hat{a}^\dagger\hat{a} + \hbar\omega_{24}\hat{b}^\dagger\hat{b} \tag{A5}$$

Each transition is driven by a laser as depicted in Fig. ??. These driving fields cause significant coupling between states, requiring the definition of an interaction Hamiltonian. Under the dipole approximation following the procedure used in [?], this interaction Hamiltonian takes the form NOTE - We do things slightly different than in the Du paper. They assume the dipole elements are all real-valued and swap $\sigma_{ij} \leftrightarrow \sigma_{ji}$ without swapping phases. This does not make any sense to me and seems like a hack or a mistake. We do not do this and end up with the addition of frequencies rather than detunings for the diagonal entries. I feel more confident with this because their method

$$\hat{H}_{Int} = -\hat{\mathbf{d}} \cdot \mathbf{E} \tag{A6}$$

$$\hat{\mathbf{d}} = \vec{d}_{12}\hat{\sigma}_{12} + \vec{d}_{13}\hat{\sigma}_{13} + \vec{d}_{24}\hat{\sigma}_{24} + \vec{d}_{34}\hat{\sigma}_{34} + h.c. \tag{A7}$$

$$\mathbf{E} = \vec{E}_I + \vec{E}_s + \vec{E}_i + \vec{E}_{II} \tag{A8}$$

where $\vec{d}_{ij}$ is the dipole operator between states i and j, the strong pump fields $\vec{E}_I$ and $\vec{E}_{II}$ are treated as classical fields, and the weak fields $\vec{E}_s$ and $\vec{E}_i$ are treated as quantized fields. Each field can be separated into a rapidly oscillating component and a slowly-varying envelope $\vec{\varepsilon}(t)$ as

$$\vec{E}_I = \vec{\varepsilon}_I(t)e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} + c.c. \tag{A9}$$

$$\vec{E}_s = \vec{\varepsilon}_s(t)\hat{a}e^{i\vec{k}_s\cdot\vec{r}} + h.c. \tag{A10}$$

$$\vec{E}_i = \vec{\varepsilon}_i(t)\hat{b}e^{i\vec{k}_i\cdot\vec{r}} + h.c. \tag{A11}$$

$$\vec{E}_{II} = \vec{\varepsilon}_{II}(t)e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}} + c.c. \tag{A12}$$

$$\tag{A13}$$

where $\hat{a}$ is the annihilation operator for the signal field, $\hat{b}$ is that of the idler field, $\vec{k}_j$ is the wavevector for each plane wave, and $\vec{r}$ is the position at which the field is experienced by the atom.

1. Classical Field Simplification via Rotating Wave Approximation

Consider some coupled classical electric field $\vec{E}_c$ and dipole operator $\hat{\vec{d}}$

$$\vec{E}_c = \vec{\varepsilon}(e^{i\phi} + e^{-i\phi}), \qquad \phi = kz - \omega t \tag{A14}$$

$$\hat{\vec{d}} = \vec{d}_{ij}\hat{\sigma}_{ij} + \vec{d}_{ji}\hat{\sigma}_{ji} \tag{A15}$$

where we may arbitrarily take $\vec{\varepsilon} = \vec{\varepsilon}^*$ and $\vec{d}_{ij} = \vec{d}_{ji}$ (real-valued vectors). We can then expand

$$\vec{E}_c \cdot \hat{\vec{d}} = \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{\sigma}_{ij}e^{i\phi} + \hat{\sigma}_{ij}e^{-i\phi} + \hat{\sigma}_{ji}e^{i\phi} + \hat{\sigma}_{ji}e^{-i\phi}\right) \tag{A16}$$

Go to the interaction picture where we have transition operators evolve as

$$\hat{\sigma}_{ij}(t) = \hat{\sigma}_{ij}e^{-i\omega_0 t} \tag{A17}$$

$$\hat{\sigma}_{ji}(t) = \hat{\sigma}_{ji}e^{i\omega_0 t} \tag{A18}$$

where ω_0 is the atomic transition frequency. We then have

$$\vec{E}_c \cdot \hat{\vec{d}} = \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{\sigma}_{ij}e^{-i\omega_0 t}e^{i\phi} + \hat{\sigma}_{ij}e^{-i\omega_0}e^{-i\phi} + \hat{\sigma}_{ji}e^{i\omega_0 t}e^{i\phi} + \hat{\sigma}_{ji}e^{i\omega_0 t}e^{-i\phi}\right)$$

$$\vec{E}_c \cdot \hat{\vec{d}} = \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{\sigma}_{ij}e^{i(kz-(\omega+\omega_0)t)} + \hat{\sigma}_{ij}e^{i(-kz+(\omega-\omega_0)t)} + \hat{\sigma}_{ji}e^{i(kz-(\omega-\omega_0)t)} + \hat{\sigma}_{ji}e^{i(-kz+(\omega+\omega_0)t)}\right) \tag{A19}$$

Assuming that the driving field of frequency ω is near-resonant with the transition, we may apply the rotating wave approximation. Oscillations at frequency $\omega + \omega_0$ are so rapid in comparison to those at frequency $\omega - \omega_0$ that we may neglect them as they average to zero on optical timescales. Thus, we have

$$\vec{E}_c \cdot \hat{\vec{d}} \approx \vec{\varepsilon} \cdot \vec{d}_{ij} \left(\hat{\sigma}_{ij} e^{i(-kz+(\omega-\omega_0)t)} + \hat{\sigma}_{ji} e^{i(kz-(\omega-\omega_0)t)} \right) \tag{A20}$$

We may then go back to the Schrödinger picture of state evolution and write

$$\vec{E}_c \cdot \hat{\vec{d}} \approx \vec{\varepsilon} \cdot \vec{d}_{ij} \left(\hat{\sigma}_{ji} e^{i(kz-\omega t)} + \hat{\sigma}_{ij} e^{i(-kz+\omega t)} \right) \tag{A21}$$

2. Quantum Field Simplification via Rotating Wave Approximation

Similar to the classical field approach, consider some coupled quantized electric field $\vec{E}_q$ and dipole operator $\hat{\vec{d}}$

$$\vec{E}_q \cdot \hat{\vec{d}} = \vec{\varepsilon} \left(\hat{a} e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger e^{-i\vec{k}\cdot\vec{r}} \right) \cdot \left(\vec{d}_{ij} \hat{\sigma}_{ij} + \vec{d}_{ji} \hat{\sigma}_{ji} \right) \tag{A22}$$

$$= \vec{\varepsilon} \cdot \vec{d}_{ij} \left(\hat{a}\hat{\sigma}_{ij} e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger \hat{\sigma}_{ij} e^{-i\vec{k}\cdot\vec{r}} + \hat{a}\vec{\sigma}_{ji} e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger \vec{\sigma}_{ji} e^{-i\vec{k}\cdot\vec{r}} \right) \tag{A23}$$

We again go to the interaction picture in which we have operators evolve as

$$\hat{a}(t) = \hat{a} e^{-i\omega t} \tag{A24}$$

$$\hat{a}^\dagger(t) = \hat{a}^\dagger e^{i\omega t} \tag{A25}$$

$$\hat{\sigma}_{ij}(t) = \hat{\sigma}_{ij} e^{-i\omega_0 t} \tag{A26}$$

$$\hat{\sigma}_{ji}(t) = \hat{\sigma}_{ji} e^{i\omega_0 t} \tag{A27}$$

So we have

$$\begin{aligned} \vec{E}_q \cdot \hat{\vec{d}} &= \vec{\varepsilon} \cdot \vec{d}_{ij} \left(\hat{a} e^{-i\omega t} e^{-i\omega_0 t} \hat{\sigma}_{ij} e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger e^{i\omega t} e^{-i\omega_0 t} \hat{\sigma}_{ij} e^{-i\vec{k}\cdot\vec{r}} + \hat{a} e^{-i\omega t} e^{i\omega_0 t} \vec{\sigma}_{ji} e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger e^{i\omega t} e^{i\omega_0 t} \vec{\sigma}_{ji} e^{-i\vec{k}\cdot\vec{r}} \right) \\ &= \vec{\varepsilon} \cdot \vec{d}_{ij} \left(\hat{a} e^{-i(\omega+\omega_0)t} \hat{\sigma}_{ij} e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger e^{-i(\omega-\omega_0)t} \hat{\sigma}_{ij} e^{-i\vec{k}\cdot\vec{r}} + \hat{a} e^{-i(\omega-\omega_0)t} \vec{\sigma}_{ji} e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger e^{i(\omega+\omega_0)t} \vec{\sigma}_{ji} e^{-i\vec{k}\cdot\vec{r}} \right) \end{aligned} \tag{A28}$$

Then again by the rotating wave approximation we have

$$\vec{E}_q \cdot \hat{\vec{d}} = \vec{\varepsilon} \cdot \vec{d}_{ij} \left(\hat{a}^\dagger e^{-i(\omega-\omega_0)t} \hat{\sigma}_{ij} e^{-i\vec{k}\cdot\vec{r}} + \hat{a} e^{-i(\omega-\omega_0)t} \vec{\sigma}_{ji} e^{i\vec{k}\cdot\vec{r}} \right) \tag{A29}$$

Going back to Schrödinger picture then yields

$$\vec{E}_q \cdot \hat{\vec{d}} = \vec{\varepsilon} \cdot \vec{d}_{ij} \left(\hat{a}^\dagger \hat{\sigma}_{ij} e^{-i\vec{k}\cdot\vec{r}} + \hat{a}\vec{\sigma}_{ji} e^{i\vec{k}\cdot\vec{r}} \right) \tag{A30}$$

$$\tag{A31}$$

3. Hamiltonian Simplification

Putting this all together, we obtain our interaction Hamiltonian as

$$\begin{aligned} \hat{H}_{Int} = &\left(\vec{d}_{12} \cdot \vec{\varepsilon}_I(t) \hat{\sigma}_{12} e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} + c.c. \right) \\ &+ \left(\vec{d}_{13} \cdot \vec{\varepsilon}_s(t) \hat{\sigma}_{13} \hat{a}^\dagger e^{i\vec{k}_s\cdot\vec{r}} + h.c. \right) \\ &+ \left(\vec{d}_{24} \cdot \vec{\varepsilon}_i(t) \hat{\sigma}_{24} \hat{b}^\dagger e^{i\vec{k}_i\cdot\vec{r}} + h.c. \right) \\ &+ \left(\vec{d}_{34} \cdot \vec{\varepsilon}_{II}(t) \hat{\sigma}_{34} e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}} + c.c. \right) \end{aligned} \tag{A32}$$

The Rabi frequency (scaled by a factor of 2 here for simplicity) is the rate at which probability amplitudes between states oscillates in the presence of an electromagnetic field [**?**] and is defined as

$$\Omega = \frac{\vec{d} \cdot \vec{\varepsilon}}{\hbar} \tag{A33}$$

so we have TODO - Quantum internet paper has g instead of Ω for weak field-atom coupling. Should we have this instead?

$$\begin{aligned}\hat{H}_{Int} = &\left(\hbar\Omega_I(t)\hat{\sigma}_{12}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + c.c.\right) \\ &+ \left(\hbar\Omega_s(t)\hat{\sigma}_{13}\hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + h.c.\right) \\ &+ \left(\hbar\Omega_i(t)\hat{\sigma}_{24}\hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} + h.c.\right) \\ &+ \left(\hbar\Omega_{II}(t)\hat{\sigma}_{34}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + c.c.\right)\end{aligned} \tag{A34}$$

where we can notice the utility of the slowly-varying envelop approximation as the Rabi oscillations change on the timescale of the envelope rather than the rapidly oscillating carrier field. This is analogous to the logic of the rotating wave approximation where the carrier field oscillates so rapidly that we can take its average effect on the dipole to be zero. This allows us to approximately separate the Rabi frequencies from the field frequencies in the above. TODO - Check this line of reasoning with MacRae.

Combining this with $\hat{H}_S$ we obtain a Hamiltonian coupling atomic states with the weak signal and idler fields as

$$\hat{H} = \hat{H}_S + \hat{H}_{Int} \tag{A35}$$

$$\begin{aligned}&= \hbar\omega_1\hat{\sigma}_{11} + \hbar\omega_2\hat{\sigma}_{22} + \hbar\omega_3\hat{\sigma}_{33} + \hbar\omega_4\hat{\sigma}_{44} \\ &\quad + \hbar\omega_{13}\hat{a}^\dagger\hat{a} + \hbar\omega_{24}\hat{b}^\dagger\hat{b} \\ &\quad + \left(\hbar\Omega_I(t)\hat{\sigma}_{12}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + c.c.\right) \\ &\quad + \left(\hbar\Omega_s(t)\hat{\sigma}_{13}\hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + h.c.\right) \\ &\quad + \left(\hbar\Omega_i(t)\hat{\sigma}_{24}\hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} + h.c.\right) \\ &\quad + \left(\hbar\Omega_{II}(t)\hat{\sigma}_{34}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + c.c.\right)\end{aligned} \tag{A36}$$

Then define the unitary transformation which rotates with state $|1\rangle$ as reference

$$\hat{U}_{atom} = \hat{\sigma}_{11} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44} \tag{A37}$$

and the unitary transform rotating at the driving field frequencies which acts on the Fock photon bases

$$\hat{U}_{photon} = e^{-i(\omega_s\hat{a}^\dagger\hat{a} + \omega_i\hat{b}^\dagger\hat{b})t} \tag{A38}$$

we then have the total unitary transformation of the coupled system as

$$\hat{U} = \hat{U}_{atom} \otimes \hat{U}_{photon} = \left(\hat{\sigma}_{11} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \otimes e^{-i(\omega_s\hat{a}^\dagger\hat{a} + \omega_i\hat{b}^\dagger\hat{b})t}. \tag{A39}$$

The total Hamiltonian can then be unitarily transformed as

$$\tilde{H} = \hat{U}^\dagger\hat{H}\hat{U} + i\hbar(\partial_t\hat{U}^\dagger)\hat{U}. \tag{A40}$$

Since the bases are independent in this transformation, we may compute the corresponding components independently

$$\hat{U}_{atom} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} & 0 & 0 \\ 0 & 0 & e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}} & 0 \\ 0 & 0 & 0 & e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}} \end{pmatrix} \tag{A41}$$

$$\begin{aligned}\partial_t \hat{U}^\dagger_{atom} &= \partial_t \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} & 0 & 0 \\ 0 & 0 & e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}} & 0 \\ 0 & 0 & 0 & e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}} \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & i\omega_I U^*_{22} & 0 & 0 \\ 0 & 0 & i\omega_s U^*_{33} & 0 \\ 0 & 0 & 0 & i(\omega_{II}+\omega_s)U^*_{44} \end{pmatrix}\end{aligned}$$

$$\begin{aligned} i\hbar(\partial_t \hat{U}^\dagger_{atom})\hat{U}_{atom} &= i\hbar \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & i\omega_I U^*_{22}U_{22} & 0 & 0 \\ 0 & 0 & i\omega_s U^*_{33}U_{33} & 0 \\ 0 & 0 & 0 & i(\omega_{II}+\omega_s)U^*_{44}U_{44} \end{pmatrix} \\ &= \hbar \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -\omega_I & 0 & 0 \\ 0 & 0 & -\omega_s & 0 \\ 0 & 0 & 0 & -(\omega_{II}+\omega_s) \end{pmatrix} \\ &= -\hbar\omega_I\hat{\sigma}_{22} - \hbar\omega_s\hat{\sigma}_{33} - \hbar(\omega_{II}+\omega_s)\hat{\sigma}_{44} \end{aligned} \tag{A42}$$

$$\partial_t \hat{U}^\dagger_{photon} = i(\omega_s\hat{a}^\dagger\hat{a} + \omega_i\hat{b}^\dagger\hat{b})e^{i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t}$$

$$\begin{aligned} i\hbar(\partial_t\hat{U}^\dagger_{photon})\hat{U}_{photon} &= i^2\hbar(\omega_s\hat{a}^\dagger\hat{a} + \omega_i\hat{b}^\dagger\hat{b})e^{i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t}e^{-i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t} \\ &= -\hbar(\omega_s\hat{a}^\dagger\hat{a} + \omega_i\hat{b}^\dagger\hat{b}) \end{aligned} \tag{A43}$$

$$i\hbar(\partial_t\hat{U}^\dagger)\hat{U} = -\hbar\omega_I\hat{\sigma}_{22} - \hbar\omega_s\hat{\sigma}_{33} - \hbar(\omega_{II}+\omega_s)\hat{\sigma}_{44} - \hbar\omega_s\hat{a}^\dagger\hat{a} - \hbar\omega_i\hat{b}^\dagger\hat{b} \tag{A44}$$

Then expanding the other term we have

$$\begin{aligned} \hat{U}^\dagger\hat{H}\hat{U} = &\left(\hat{\sigma}_{11} + e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \otimes e^{i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t} \\ &\cdot \Big[\hbar\omega_1\hat{\sigma}_{11} + \hbar\omega_2\hat{\sigma}_{22} + \hbar\omega_3\hat{\sigma}_{33} + \hbar\omega_4\hat{\sigma}_{44} \\ &\quad + \hbar\omega_{13}\hat{a}^\dagger\hat{a} + \hbar\omega_{24}\hat{b}^\dagger\hat{b} \\ &\quad + \hbar\left(\Omega_I(t)\hat{\sigma}_{12}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + \Omega_I^*(t)\hat{\sigma}_{21}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\right) \\ &\quad + \hbar\left(\Omega_s(t)\hat{\sigma}_{13}\hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + \Omega_s^*(t)\hat{\sigma}_{31}\hat{a}e^{i\vec{k}_s\cdot\vec{r}}\right) \\ &\quad + \hbar\left(\Omega_i(t)\hat{\sigma}_{24}\hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} + \Omega_i^*(t)\hat{\sigma}_{42}\hat{b}e^{i\vec{k}_i\cdot\vec{r}}\right) \\ &\quad + \hbar\left(\Omega_{II}(t)\hat{\sigma}_{34}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + \Omega_{II}^*(t)\hat{\sigma}_{43}e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}}\right)\Big] \\ &\cdot \left(\hat{\sigma}_{11} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \otimes e^{-i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t} \end{aligned} \tag{A45}$$

Since $\hat{U}_{atom}$ commutes with $\hat{U}_{photon}$, and $\hat{U}_{atom}$ commutes with $\hat{H}_{photon}$ such that

$$\hat{U}^{\dagger}_{atom}\hat{H}_{photon}\hat{U}_{atom} = \hat{U}^{\dagger}_{atom}\hat{U}_{atom}\hat{H}_{photon} = \hat{H}_{photon} \tag{A46}$$

we have

$$\begin{aligned}
\frac{\hat{U}^{\dagger}\hat{H}\hat{U}}{\hbar} = {} & e^{i(\omega_s\hat{a}^{\dagger}\hat{a}+\omega_i\hat{b}^{\dagger}\hat{b})t} \\
& \otimes \left(\hat{\sigma}_{11} + e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \\
& \cdot \Big[\omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
& \quad + \omega_{13}\hat{a}^{\dagger}\hat{a} + \omega_{24}\hat{b}^{\dagger}\hat{b} \\
& \quad + \left(\Omega_I(t)\hat{\sigma}_{12}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + \Omega_I^*(t)\hat{\sigma}_{21}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\right) \\
& \quad + \left(\Omega_s(t)\hat{\sigma}_{13}\hat{a}^{\dagger}e^{-i\vec{k}_s\cdot\vec{r}} + \Omega_s^*(t)\hat{\sigma}_{31}\hat{a}e^{i\vec{k}_s\cdot\vec{r}}\right) \\
& \quad + \left(\Omega_i(t)\hat{\sigma}_{24}\hat{b}^{\dagger}e^{-i\vec{k}_i\cdot\vec{r}} + \Omega_i^*(t)\hat{\sigma}_{42}\hat{b}e^{i\vec{k}_i\cdot\vec{r}}\right) \\
& \quad + \left(\Omega_{II}(t)\hat{\sigma}_{34}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + \Omega_{II}^*(t)\hat{\sigma}_{43}e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}}\right)\Big] \\
& \cdot \left(\hat{\sigma}_{11} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \\
& \otimes e^{-i(\omega_s\hat{a}^{\dagger}\hat{a}+\omega_i\hat{b}^{\dagger}\hat{b})t} \\
& + e^{i(\omega_s\hat{a}^{\dagger}\hat{a}+\omega_i\hat{b}^{\dagger}\hat{b})t}\left(\omega_{13}\hat{a}^{\dagger}\hat{a} + \omega_{24}\hat{b}^{\dagger}\hat{b}\right)e^{-i(\omega_s\hat{a}^{\dagger}\hat{a}+\omega_i\hat{b}^{\dagger}\hat{b})t}
\end{aligned} \tag{A47}$$

Separate the terms as

$$\begin{aligned}
\hat{A} = {} & \left(\hat{\sigma}_{11} + e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \\
& \cdot \Big[\omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
& \quad + \omega_{13}\hat{a}^{\dagger}\hat{a} + \omega_{24}\hat{b}^{\dagger}\hat{b} \\
& \quad + \left(\Omega_I(t)\hat{\sigma}_{12}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + \Omega_I^*(t)\hat{\sigma}_{21}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\right) \\
& \quad + \left(\Omega_s(t)\hat{\sigma}_{13}\hat{a}^{\dagger}e^{-i\vec{k}_s\cdot\vec{r}} + \Omega_s^*(t)\hat{\sigma}_{31}\hat{a}e^{i\vec{k}_s\cdot\vec{r}}\right) \\
& \quad + \left(\Omega_i(t)\hat{\sigma}_{24}\hat{b}^{\dagger}e^{-i\vec{k}_i\cdot\vec{r}} + \Omega_i^*(t)\hat{\sigma}_{42}\hat{b}e^{i\vec{k}_i\cdot\vec{r}}\right) \\
& \quad + \left(\Omega_{II}(t)\hat{\sigma}_{34}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + \Omega_{II}^*(t)\hat{\sigma}_{43}e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}}\right)\Big] \\
& \cdot \left(\hat{\sigma}_{11} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right)
\end{aligned} \tag{A48}$$

$$\hat{B} = e^{i(\omega_s\hat{a}^{\dagger}\hat{a}+\omega_i\hat{b}^{\dagger}\hat{b})t}\hat{A}e^{-i(\omega_s\hat{a}^{\dagger}\hat{a}+\omega_i\hat{b}^{\dagger}\hat{b})t} \tag{A49}$$

$$\hat{C} = \left(e^{i(\omega_s\hat{a}^{\dagger}\hat{a}+\omega_i\hat{b}^{\dagger}\hat{b})t}\right)\left(\omega_{13}\hat{a}^{\dagger}\hat{a} + \omega_{24}\hat{b}^{\dagger}\hat{b}\right)\left(e^{-i(\omega_s\hat{a}^{\dagger}\hat{a}+\omega_i\hat{b}^{\dagger}\hat{b})t}\right) \tag{A50}$$

Let us compute $\hat{A}$ using the facts that

$$\hat{\sigma}_{ij}\hat{\sigma}_{kl} = \hat{\sigma}_{il}\delta_{jk}, \tag{A51}$$

$$\hat{\sigma}_{ij}\hat{\sigma}_{jk}\hat{\sigma}_{kl} = \hat{\sigma}_{il}, \tag{A52}$$

and

$$[\hat{\sigma}_{ij}, \hat{a}] = [\hat{\sigma}_{ij}, \hat{a}^\dagger] = [\hat{\sigma}_{ij}, \hat{b}] = [\hat{\sigma}_{ij}, \hat{b}^\dagger] = 0. \tag{A53}$$

First notice that

$$\begin{aligned}
&\Big(\hat{\sigma}_{11} + e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\Big)\\
&\cdot\Big[\omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44}\Big]\\
&\cdot\Big(\hat{\sigma}_{11} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\Big) \qquad \text{(A54)}\\
&\quad = \Big[\omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44}\Big]
\end{aligned} \tag{A55}$$

So we have

$$\begin{aligned}
\hat{A} &= \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44}\\
&\quad + \Big(\hat{\sigma}_{11} + e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\Big)\\
&\quad\cdot\Big[\Big(\Omega_I(t)\hat{\sigma}_{12}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + \Omega_I^*(t)\hat{\sigma}_{21}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\Big)\\
&\quad\ + \Big(\Omega_s(t)\hat{\sigma}_{13}\hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + \Omega_s^*(t)\hat{\sigma}_{31}\hat{a}e^{i\vec{k}_s\cdot\vec{r}}\Big)\\
&\quad\ + \Big(\Omega_i(t)\hat{\sigma}_{24}\hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} + \Omega_i^*(t)\hat{\sigma}_{42}\hat{b}e^{i\vec{k}_i\cdot\vec{r}}\Big)\\
&\quad\ + \Big(\Omega_{II}(t)\hat{\sigma}_{34}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + \Omega_{II}^*(t)\hat{\sigma}_{43}e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}}\Big)\Big]\\
&\quad\cdot\Big(\hat{\sigma}_{11} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\Big)\\[1em]
&= \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44}\\
&\quad + \Big(\Omega_I(t)\hat{\sigma}_{11}\hat{\sigma}_{12}\hat{\sigma}_{22}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + \Omega_I^*(t)e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22}\hat{\sigma}_{21}\hat{\sigma}_{11}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\Big)\\
&\quad + \Big(\Omega_s(t)\hat{\sigma}_{11}\hat{\sigma}_{13}\hat{\sigma}_{33}e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + \Omega_s^*(t)e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33}\hat{\sigma}_{31}\hat{\sigma}_{11}\hat{a}e^{i\vec{k}_s\cdot\vec{r}}\Big)\\
&\quad + \Big(\Omega_i(t)e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{22}\hat{\sigma}_{24}\hat{\sigma}_{44}e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}}\\
&\quad\quad + \Omega_i^*(t)e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\hat{\sigma}_{42}\hat{\sigma}_{22}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{b}e^{i\vec{k}_i\cdot\vec{r}}\Big)\\
&\quad + \Big(\Omega_{II}(t)e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{33}\hat{\sigma}_{34}\hat{\sigma}_{44}e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}}\\
&\quad\quad + \Omega_{II}^*(t)e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\hat{\sigma}_{43}\hat{\sigma}_{33}e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}}\Big)\\[1em]
&= \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44}\\
&\quad + (\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21})\\
&\quad + \big(\Omega_s(t)e^{-i\omega_s t}\hat{\sigma}_{13}\hat{a}^\dagger + \Omega_s^*(t)e^{i\omega_s t}\hat{\sigma}_{31}\hat{a}\big)\\
&\quad + \Big(\Omega_i(t)e^{i(\omega_I - \omega_{II} - \omega_s)t - i(\vec{k}_I + \vec{k}_i - \vec{k}_{II} - \vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{24}\hat{b}^\dagger + \Omega_i^*(t)e^{-i(\omega_{II} + \omega_s - \omega_I)t + i(\vec{k}_I + \vec{k}_i - \vec{k}_{II} - \vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{42}\hat{b}\Big)\\
&\quad + (\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43})
\end{aligned} \tag{A56}$$

We can then impose the energy conservation condition and phase match condition (in that order as follows)

$$\omega_I + \omega_i = \omega_s + \omega_{II} \tag{A57}$$
$$k_I + k_i = k_s + k_{II} \tag{A58}$$

$$\begin{aligned}\hat{A} = {} & \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\ & + (\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21}) \\ & + \left(\Omega_s(t)e^{-i\omega_s t}\hat{\sigma}_{13}\hat{a}^\dagger + \Omega_s^*(t)e^{i\omega_s t}\hat{\sigma}_{31}\hat{a}\right) \\ & + \left(\Omega_i(t)e^{-i\omega_i t}\hat{\sigma}_{24}\hat{b}^\dagger + \Omega_i^*(t)e^{i\omega_i t}\hat{\sigma}_{42}\hat{b}\right) \\ & + (\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43})\end{aligned} \tag{A59}$$

Now we may compute $\hat{B}$ from eq. (**??**) as

$$\begin{aligned}\hat{B} &= e^{i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t}\hat{A}e^{-i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t} \\ &= e^{i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t} \\ &\quad \cdot \Big[\omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\ &\quad + (\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21}) \\ &\quad + \left(\Omega_s(t)e^{-i\omega_s t}\hat{\sigma}_{13}\hat{a}^\dagger + \Omega_s^*(t)e^{i\omega_s t}\hat{\sigma}_{31}\hat{a}\right) \\ &\quad + \left(\Omega_i(t)e^{-i\omega_i t}\hat{\sigma}_{24}\hat{b}^\dagger + \Omega_i^*(t)e^{i\omega_i t}\hat{\sigma}_{42}\hat{b}\right) \\ &\quad + (\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43})\Big] \\ &\quad \cdot e^{-i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t}\end{aligned} \tag{A60}$$

which, via commutation relations, simplifies to

$$\begin{aligned}\hat{B} = {} & \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\ & + (\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21}) \\ & + (\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43}) \\ & + e^{i\omega_s\hat{a}^\dagger\hat{a}t}\left(\Omega_s(t)e^{-i\omega_s t}\hat{\sigma}_{13}\hat{a} + \Omega_s^*(t)e^{i\omega_s t}\hat{\sigma}_{31}\hat{a}^\dagger\right)e^{-i\omega_s\hat{a}^\dagger\hat{a}t} \\ & + e^{i\omega_i\hat{b}^\dagger\hat{b}t}\left(\Omega_i(t)e^{-i\omega_i t}\hat{\sigma}_{24}\hat{b} + \Omega_i^*(t)e^{i\omega_i t}\hat{\sigma}_{42}\hat{b}^\dagger\right)e^{-i\omega_i\hat{b}^\dagger\hat{b}t}\end{aligned} \tag{A61}$$

We must then compute

$$e^{i\omega_s\hat{a}^\dagger\hat{a}t}\hat{a}e^{-i\omega_s\hat{a}^\dagger\hat{a}t} \tag{A62}$$

which we can do via the Baker-Campbell-Hausdorff formula

$$e^{\hat{X}}\hat{Y}e^{-\hat{X}} = \sum_{n=0}^{\infty}\frac{[X,Y]_n}{n!} = Y + [X,Y] + \frac{1}{2!}[X,[X,Y]] + \cdots \tag{A63}$$

where

$$[X,Y]_n = \underbrace{[X,[X,\ldots,[X,[X}_{\text{n times}},Y]],\ldots]] \tag{A64}$$

$$e^{i\omega_s \hat{a}^\dagger \hat{a} t}\hat{a}e^{-i\omega_s \hat{a}^\dagger \hat{a} t} = \sum_{n=0}^{\infty} \frac{[i\omega_s \hat{a}^\dagger \hat{a} t, \hat{a}]_n}{n!}$$

$$\begin{aligned}
e^{i\omega_s \hat{a}^\dagger \hat{a} t}\hat{a}e^{-i\omega_s \hat{a}^\dagger \hat{a} t} &= \sum_{n=0}^{\infty} \frac{[i\omega_s \hat{a}^\dagger \hat{a} t, \hat{a}]_n}{n!} \\
&= \hat{a} + [i\omega_s \hat{a}^\dagger \hat{a} t, \hat{a}] + \frac{1}{2!}[i\omega_s \hat{a}^\dagger \hat{a} t, [i\omega_s \hat{a}^\dagger \hat{a} t, \hat{a}]] + \frac{1}{3!}[i\omega_s \hat{a}^\dagger \hat{a} t, [i\omega_s \hat{a}^\dagger \hat{a} t, [i\omega_s \hat{a}^\dagger \hat{a} t, \hat{a}]]] + \cdots \\
&= \hat{a} + i\omega_s t[\hat{a}^\dagger \hat{a}, \hat{a}] + \frac{(i\omega_s t)^2}{2!}[\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, \hat{a}]] + \frac{(i\omega_s t)^3}{3!}[\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, \hat{a}]]] + \cdots
\end{aligned} \tag{A65}$$

To move forward we must then compute the commutators

$$[\hat{a}^\dagger \hat{a}, \hat{a}] = \hat{a}^\dagger \hat{a}\hat{a} - \hat{a}\hat{a}^\dagger \hat{a}$$

use the commutation relation

$$\begin{aligned}
[\hat{a}, \hat{a}^\dagger] &= \hat{a}\hat{a}^\dagger - \hat{a}^\dagger \hat{a} = 1 \\
\hat{a}\hat{a}^\dagger &= 1 + \hat{a}^\dagger \hat{a}
\end{aligned} \tag{A66}$$

to obtain

$$\begin{aligned}
[\hat{a}^\dagger \hat{a}, \hat{a}] &= \hat{a}^\dagger \hat{a}\hat{a} - (1 + \hat{a}^\dagger \hat{a})\hat{a} \\
&= \hat{a}^\dagger \hat{a}\hat{a} - \hat{a} - \hat{a}^\dagger \hat{a}\hat{a} \\
&= -\hat{a}
\end{aligned} \tag{A67}$$

Then we have that

$$\begin{aligned}
[\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, \hat{a}]] &= [\hat{a}^\dagger \hat{a}, -\hat{a}] \\
&= -[\hat{a}^\dagger \hat{a}, \hat{a}] \\
&= \hat{a}
\end{aligned} \tag{A68}$$

Assume this pattern holds until the k^{th} application

$$[\hat{a}^\dagger \hat{a}, \hat{a}]_k = (-1)^k \hat{a} \qquad \text{(Induction Hypothesis)} \tag{A69}$$

then consider the $(k+1)^{th}$ step

$$\begin{aligned}
[\hat{a}^\dagger \hat{a}, \hat{a}]_{k+1} &= [\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, \hat{a}]_k] \\
&= [\hat{a}^\dagger \hat{a}, (-1)^k \hat{a}] \\
&= (-1)^k [\hat{a}^\dagger \hat{a}, \hat{a}] \\
&= (-1)^k (-\hat{a}) \\
&= (-1)^{k+1} \hat{a}
\end{aligned}$$

Thus, by induction we have that

$$[\hat{a}^\dagger \hat{a}, \hat{a}]_n = (-1)^n \hat{a} \tag{A70}$$

So we may write

$$\begin{aligned}
e^{i\omega_s \hat{a}^\dagger \hat{a} t}\hat{a}e^{-i\omega_s \hat{a}^\dagger \hat{a} t} &= \hat{a} + i\omega_s t[\hat{a}^\dagger \hat{a}, \hat{a}] + \frac{(i\omega_s t)^2}{2!}[\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, \hat{a}]] + \frac{(i\omega_s t)^3}{3!}[\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, \hat{a}]]] + \cdots \\
&= \hat{a} + i\omega_s t(-\hat{a}) + \frac{(i\omega_s t)^2}{2!}(\hat{a}) + \frac{(i\omega_s t)^3}{3!}(-\hat{a}) + \cdots \\
&= \hat{a}\left(1 - i\omega_s t + \frac{(i\omega_s t)^2}{2!} - \frac{(i\omega_s t)^3}{3!} + \cdots\right) \\
&= \hat{a}\left(1 + (-i\omega_s t) + \frac{(-i\omega_s t)^2}{2!} + \frac{(-i\omega_s t)^3}{3!} + \cdots\right) \\
&= \hat{a}\sum_{n=0}^{\infty} \frac{(-i\omega_s t)^n}{n!} \\
&= \hat{a}e^{-i\omega_s t}
\end{aligned} \tag{A71}$$

Then we can find

$$\begin{aligned}
&\left(e^{i\omega_s \hat{a}^\dagger \hat{a} t}\hat{a}e^{-i\omega_s \hat{a}^\dagger \hat{a} t}\right)^\dagger = \left(\hat{a}e^{-i\omega_s t}\right)^\dagger \\
&\left(e^{-i\omega_s \hat{a}^\dagger \hat{a} t}\right)^\dagger \hat{a}^\dagger \left(e^{i\omega_s \hat{a}^\dagger \hat{a} t}\right)^\dagger = \hat{a}^\dagger e^{i\omega_s t} \\
&e^{i\omega_s \left(\hat{a}^\dagger \hat{a}\right)^\dagger t}\hat{a}^\dagger e^{i\omega_s \left(\hat{a}^\dagger \hat{a}\right)^\dagger t} = \hat{a}^\dagger e^{i\omega_s t} \\
&e^{i\omega_s \hat{a}^\dagger \left(\hat{a}^\dagger\right)^\dagger t}\hat{a}^\dagger e^{i\omega_s \hat{a}^\dagger \left(\hat{a}^\dagger\right)^\dagger \hat{a} t} = \hat{a}^\dagger e^{i\omega_s t} \\
&e^{i\omega_s \hat{a}^\dagger \hat{a} t}\hat{a}^\dagger e^{-i\omega_s \hat{a}^\dagger \hat{a} t} = \hat{a}^\dagger e^{i\omega_s t}
\end{aligned} \tag{A72}$$

Similarly

$$e^{i\omega_i \hat{b}^\dagger \hat{b} t}\hat{b}e^{-i\omega_i \hat{b}^\dagger \hat{b} t} = \hat{b}e^{-i\omega_i t} \tag{A73}$$

$$e^{i\omega_i \hat{b}^\dagger \hat{b} t}\hat{b}^\dagger e^{-i\omega_i \hat{b}^\dagger \hat{b} t} = \hat{b}^\dagger e^{i\omega_i t} \tag{A74}$$

So we can simplify eq. (**??**) to

$$\begin{aligned}
\hat{B} = {} & \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
& + \left(\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21}\right) \\
& + \left(\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43}\right) \\
& + \left(\Omega_s(t)e^{-i\omega_s t}\hat{\sigma}_{13}\hat{a}^\dagger e^{i\omega_s t} + \Omega_s^*(t)e^{i\omega_s t}\hat{\sigma}_{31}\hat{a}e^{-i\omega_s t}\right) \\
& + \left(\Omega_i(t)e^{-i\omega_i t}\hat{\sigma}_{24}\hat{b}^\dagger e^{i\omega_i t} + \Omega_i^*(t)e^{i\omega_i t}\hat{\sigma}_{42}\hat{b}e^{-i\omega_i t}\right)
\end{aligned}$$

$$\begin{aligned}
\hat{B} = {} & \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
& + \left(\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21}\right) \\
& + \left(\Omega_s(t)\hat{a}^\dagger\hat{\sigma}_{13} + \Omega_s^*(t)\hat{a}\hat{\sigma}_{31}\right) \\
& + \left(\Omega_i(t)\hat{b}^\dagger\hat{\sigma}_{24} + \Omega_i^*(t)\hat{b}\hat{\sigma}_{42}\right) \\
& + \left(\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43}\right)
\end{aligned} \tag{A75}$$

$$\hat{B} = \begin{pmatrix}
\omega_1 & \Omega_I(t) & \Omega_s(t)\hat{a}^\dagger & 0 \\
\Omega_I^*(t) & \omega_2 & 0 & \Omega_i(t)\hat{b}^\dagger \\
\Omega_s^*(t)\hat{a} & 0 & \omega_3 & \Omega_{II}(t) \\
0 & \Omega_i^*(t)\hat{b} & \Omega_{II}^*(t) & \omega_4
\end{pmatrix} \tag{A76}$$

Now consider the next component from eq. (**??**)

$$\begin{aligned}
\hat{C} &= \left(e^{i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t}\right)\left(\omega_{13}\hat{a}^\dagger\hat{a} + \omega_{24}\hat{b}^\dagger\hat{b}\right)\left(e^{-i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t}\right) \\
&= \omega_{13}e^{i\omega_s \hat{a}^\dagger \hat{a} t}\hat{a}^\dagger\hat{a}e^{-i\omega_s \hat{a}^\dagger \hat{a}} + \omega_{24}e^{\omega_i \hat{b}^\dagger \hat{b} t}\hat{b}^\dagger\hat{b}e^{-\omega_i \hat{b}^\dagger \hat{b} t}
\end{aligned} \tag{A77}$$

using the commutation relation for the distinct fields

$$\left[\hat{a}^\dagger\hat{a}, \hat{b}^\dagger\hat{b}\right] = 0 \tag{A78}$$

Then we may expand the exponentials as

$$\begin{aligned}
e^{i\omega\hat{a}^\dagger\hat{a}t}\hat{a}^\dagger\hat{a}e^{-i\omega\hat{a}^\dagger\hat{a}t} &= \left[\sum_{n=0}^{\infty}\frac{\left(i\omega\hat{a}^\dagger\hat{a}t\right)^n}{n!}\right]\hat{a}^\dagger\hat{a}\left[\sum_{n=0}^{\infty}\frac{\left(-i\omega\hat{a}^\dagger\hat{a}t\right)^n}{n!}\right] \\
&= \left[\sum_{n=0}^{\infty}\frac{(i\omega t)^n\left(\hat{a}^\dagger\hat{a}\right)^{n+1}}{n!}\right]\left[\sum_{n=0}^{\infty}\frac{\left(-i\omega\hat{a}^\dagger\hat{a}t\right)^n}{n!}\right] \\
&= \hat{a}^\dagger\hat{a}\left[\sum_{n=0}^{\infty}\frac{\left(i\omega\hat{a}^\dagger\hat{a}t\right)^n}{n!}\right]\left[\sum_{n=0}^{\infty}\frac{\left(-i\omega\hat{a}^\dagger\hat{a}t\right)^n}{n!}\right] \\
&= \hat{a}^\dagger\hat{a}e^{i\omega\hat{a}^\dagger\hat{a}t}e^{-i\omega\hat{a}^\dagger\hat{a}t} \\
&= \hat{a}^\dagger\hat{a}
\end{aligned} \tag{A79}$$

which follows similarly for

$$e^{i\omega\hat{b}^\dagger\hat{b}t}\hat{b}^\dagger\hat{b}e^{-i\omega\hat{b}^\dagger\hat{b}t} = \hat{b}^\dagger\hat{b} \tag{A80}$$

yielding

$$\hat{C} = \omega_{13}\hat{a}^\dagger\hat{a} + \omega_{24}\hat{b}^\dagger\hat{b} \tag{A81}$$

Thus, we have

$$\begin{aligned}
\frac{\hat{U}^\dagger\hat{H}\hat{U}}{\hbar} &= \hat{B} + \hat{C} \\
&= \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
&\quad + \left(\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21}\right) \\
&\quad + \left(\Omega_s(t)\hat{a}^\dagger\hat{\sigma}_{13} + \Omega_s^*(t)\hat{a}\hat{\sigma}_{31}\right) \\
&\quad + \left(\Omega_i(t)\hat{b}^\dagger\hat{\sigma}_{24} + \Omega_i^*(t)\hat{b}\hat{\sigma}_{42}\right) \\
&\quad + \left(\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43}\right) \\
&\quad + \omega_{13}\hat{a}^\dagger\hat{a} + \omega_{24}\hat{b}^\dagger\hat{b}
\end{aligned} \tag{A82}$$

Then we have the unitary transformation as

$$\begin{aligned}
\tilde{H} &= \hat{U}^\dagger\hat{H}\hat{U} + i\hbar(\partial_t\hat{U}^\dagger)\hat{U} \\
&= \hbar\Big[\omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
&\qquad + \left(\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21}\right) \\
&\qquad + \left(\Omega_s(t)\hat{a}^\dagger\hat{\sigma}_{13} + \Omega_s^*(t)\hat{a}\hat{\sigma}_{31}\right) \\
&\qquad + \left(\Omega_i(t)\hat{b}^\dagger\hat{\sigma}_{24} + \Omega_i^*(t)\hat{b}\hat{\sigma}_{42}\right) \\
&\qquad + \left(\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43}\right) \\
&\qquad + \omega_{13}\hat{a}^\dagger\hat{a} + \omega_{24}\hat{b}^\dagger\hat{b}\Big] \\
&\quad + \hbar\Big[-\omega_I\hat{\sigma}_{22} - \omega_s\hat{\sigma}_{33} - (\omega_{II} + \omega_s)\hat{\sigma}_{44} - \omega_s\hat{a}^\dagger\hat{a} - \omega_i\hat{b}^\dagger\hat{b}\Big]
\end{aligned}$$

$$\begin{aligned}\tilde{H} = \hbar\Big[&\omega_1\hat{\sigma}_{11} + (\omega_2 - \omega_I)\hat{\sigma}_{22} + (\omega_3 - \omega_s)\hat{\sigma}_{33} + (\omega_4 - \omega_{II} + \omega_s)\hat{\sigma}_{44} \\ &+ (\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21}) \\ &+ \left(\Omega_s(t)\hat{a}^\dagger\hat{\sigma}_{13} + \Omega_s^*(t)\hat{a}\hat{\sigma}_{31}\right) \\ &+ \left(\Omega_i(t)\hat{b}^\dagger\hat{\sigma}_{24} + \Omega_i^*(t)\hat{b}\hat{\sigma}_{42}\right) \\ &+ (\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43}) \\ &+ (\omega_{13} - \omega_s)\hat{a}^\dagger\hat{a} + (\omega_{24} - \omega_i)\hat{b}^\dagger\hat{b}\Big]\end{aligned} \tag{A83}$$

We can then choose to arbitrarily shift the energy levels of the Hamiltonian by

$$\begin{aligned}&\tilde{H} \to \tilde{H} - \hbar\omega_1\hat{\mathbb{1}} \\ &\to \hbar\Big[(\omega_2 - \omega_1 - \omega_I)\hat{\sigma}_{22} + (\omega_3 - \omega_1 - \omega_s)\hat{\sigma}_{33} + (\omega_4 - \omega_1 - \omega_{II} - \omega_s)\hat{\sigma}_{44} \\ &+ (\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21}) \\ &+ \left(\Omega_s(t)\hat{a}^\dagger\hat{\sigma}_{13} + \Omega_s^*(t)\hat{a}\hat{\sigma}_{31}\right) \\ &+ \left(\Omega_i(t)\hat{b}^\dagger\hat{\sigma}_{24} + \Omega_i^*(t)\hat{b}\hat{\sigma}_{42}\right) \\ &+ (\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43}) \\ &+ (\omega_{13} - \omega_s)\hat{a}^\dagger\hat{a} + (\omega_{24} - \omega_i)\hat{b}^\dagger\hat{b}\Big]\end{aligned} \tag{A84}$$

Apply the relations

$$\omega_{12} = \omega_2 - \omega_1 \tag{A85}$$
$$\omega_{13} = \omega_3 - \omega_1 \tag{A86}$$
$$\omega_{24} = \omega_4 - \omega_2 \tag{A87}$$
$$\omega_{34} = \omega_4 - \omega_3 \tag{A88}$$
$$\Delta_s = \omega_{13} - \omega_s \tag{A89}$$
$$\Delta_i = \omega_{24} - \omega_i \tag{A90}$$
$$\Delta_I = \omega_{12} - \omega_I \tag{A91}$$
$$\Delta_{II} = \omega_{34} - \omega_{II} \tag{A92}$$
$$\tag{A93}$$

we have

$$\begin{aligned}\omega_{34} + \omega_{13} &= (\omega_4 - \omega_3) + (\omega_3 - \omega_1) \\ &= \omega_4 - \omega_1\end{aligned} \tag{A94}$$

$$\begin{aligned}\tilde{H} = \hbar\Big[&\Delta_I\hat{\sigma}_{22} + \Delta_s\hat{\sigma}_{33} + (\Delta_{II} + \Delta_s)\hat{\sigma}_{44} \\ &+ (\Omega_I(t)\hat{\sigma}_{12} + \Omega_I^*(t)\hat{\sigma}_{21}) \\ &+ \left(\Omega_s(t)\hat{a}^\dagger\hat{\sigma}_{13} + \Omega_s^*(t)\hat{a}\hat{\sigma}_{31}\right) \\ &+ \left(\Omega_i(t)\hat{b}^\dagger\hat{\sigma}_{24} + \Omega_i^*(t)\hat{b}\hat{\sigma}_{42}\right) \\ &+ (\Omega_{II}(t)\hat{\sigma}_{34} + \Omega_{II}^*(t)\hat{\sigma}_{43}) \\ &+ \Delta_s\hat{a}^\dagger\hat{a} + \Delta_i\hat{b}^\dagger\hat{b}\Big]\end{aligned} \tag{A95}$$

$$\tilde{H} = \hbar \begin{pmatrix} 0 & \Omega_I(t) & \Omega_s(t)\hat{a}^\dagger & 0 \\ \Omega_I^*(t) & \Delta_I & 0 & \Omega_i(t)\hat{b}^\dagger \\ \Omega_s^*(t)\hat{a} & 0 & \Delta_s & \Omega_{II}(t) \\ 0 & \Omega_i^*(t)\hat{b} & \Omega_{II}^*(t) & \Delta_{II}+\Delta_s \end{pmatrix} + \hbar\Delta_s\hat{a}^\dagger\hat{a} + \hbar\Delta_i\hat{b}^\dagger\hat{b} \tag{A96}$$

Appendix B: Maxwell-Bloch Equation Derivation

TODO - THE WORRYING DERIVATIVE GOES TO ZERO AS WE USE A STEADY STATE!! WE'RE GOOD
I am confused here. Operator time evolution in the Heisenberg picture is

$$\frac{d}{dt}A_H(t) = \frac{i}{\hbar}\left[H_H(t), A_H(t)\right] + \left(\frac{\partial A_S}{\partial t}\right)_H \tag{B1}$$

We would want to do this with the weak quantized electric fields $\hat{\vec{E}}_s(t) = \vec{\varepsilon}_s(t)\hat{a}e^{i\vec{k}_s\cdot\vec{r}} + h.c.$, so we would have

$$\left(\frac{\partial\hat{\vec{E}}_s}{\partial t}\right)_H = \left(\hat{a}e^{i\vec{k}_s\cdot\vec{r}}\frac{\partial\vec{\varepsilon}_s(t)}{\partial t} + h.c.\right)_H \tag{B2}$$

So we end up with

$$\frac{d}{dt}\hat{\vec{E}}_{H,s}(t) = \frac{i}{\hbar}\left[\tilde{H}(t), \hat{\vec{E}}_{H,s}(t)\right] + \left(\hat{a}e^{i\vec{k}_s\cdot\vec{r}}\frac{\partial\vec{\varepsilon}_s(t)}{\partial t} + h.c.\right)_H \tag{B3}$$

This then couples $(d/d_t)\hat{\vec{E}}_{H,s}(t)$ and $\partial_t\vec{\varepsilon}_s(t)$. This differs from Kupchak's approach where $\partial_t\vec{\varepsilon}_s(t)$ is not involved as he takes $\left(\partial_t\hat{\vec{E}}_s\right)_H = 0$. Am I incorrect in my equation for $\hat{\vec{E}}_s(t) = \vec{\varepsilon}_s(t)\hat{a}e^{i\vec{k}_s\cdot\vec{r}} + h.c.$? Should it not be time dependent yet? This would solve this discrepancy, and then we should be able to use Kupchak's approach.

TODO - Use the Glorieux reference to do Kupchak's pressure equations